## Order Insectivora

The six families of the order Insectivora are a taxonomic hodge-podge with little close phylogenetic relationship. A 19th century monograph on the order is available (2662) as is a review of reproductive physiology (9350).

## Order Insectivora, Family Solenodontidae, Genus *Solenodon*

The 1 kg, nocturnal solenodonts (*Solenodon cubanus*, *S. paradoxus*) from Cuba, Santa Domingo and Haiti are insectivores with antimesometrial implantation of a hemochorial, discoidal, labyrinthine placenta into a uterus with a large body and two blunt horns (176, 11853). Naked and blind young (176, 1285, 11269) suckle from two teats near the base of the tail (176, 310). The teats are also used to transport young (2930). Females have a urogenital sinus (176, 11853). Testes are abdominal and the penis retracts into the abdominal wall (176, 2662, 8967). A German monograph on *S. paradoxus* is available (7426, 7427, 7428, 7429; cf 2933).

| | | |
|---|---|---|
| **NEONATAL MASS** | | |
| *Solenodon paradoxus* | est 100 g | 2930 |
| **LITTER SIZE** | | |
| *Solenodon cubanus* | mode 1-2 | 10809 |
| *Solenodon paradoxus* | 1 emb; n1 f | 11853 |
| | 1 emb; n1 f | 176 |
| | 1 yg; n1 lt | 1285 |
| | mode 2; r1-3 | 7428 |
| | 3; n1 lt | 11269 |
| **CYCLE LENGTH** | | |
| *Solenodon paradoxus* | 9-13 d | 2930 |
| **GESTATION** | | |
| *Solenodon paradoxus* | est >50 d | 2930 |
| **LACTATION** | | |
| *Solenodon paradoxus* | wean: est 75 d | 2930 |

## Order Insectivora, Family Tenrecidae

The solitary, nocturnal tenrecs and their relatives (*Dasogale*, *Echinops* 110-250 g; *Geogale* 71 mm; *Hemicentetes* 70-170 g; *Leptogale*, *Limnogale* 50-90 g; *Microgale* 34-85 g; *Micropotamogale* 135 g; *Oryzorictes* 42-48 g; *Potamogale* 1 kg; *Setifer* 125-365 g; *Tenrec* 830-2400 g) are terrestrial, semiaquatic, or semifossorial predators on invertebrates in Madagascar and Africa. The reproductive biology of *Dasogale*, *Geogale*, *Leptogale*, *Limnogale*, and *Oryzorictes* is not known.

Male *Echinops* have an additional chromosome (3492). *Potamogale* has a simplex uterus with two nonfunctional cornu (4762), while other genera have bicornuate uteri (3218, 3855, 9236, 10484). Females have well-developed prostate and bulbourethral glands (6364, 8436). *Hemicentetes* (3218, 3857) and *Setifer* (8620) are reportedly polyestrous, although tenrecs may be induced ovulators (7984). Ovarian follicles lack antra, and fertilization occurs within the ovary (3218, 3219, 6169, 7892, 10484, 10485). Polyovular follicles are common (*Tenrec* 7890). *Hemicentetes* has no morula (3855), but *Tenrec* does (7892). The labyrinthine, hemochorial, discoidal or diffuse placenta has a circumplacental chorion, a hemophagous organ, but no consistent implantation site (1025, 3218, 3854, 3855, 3856, 4104, 4762, 4814, 7563, 7891, 10310, 10486). Placentophagia occurs (3975, 4949, 8620).

The naked neonates open their eyes at 7-9 d (*Echinops* 2942, 3844, 3975; cf 9-15 d 4949, 9138), 7-10 d (*Hemicentetes* 2934, 2942, 3454, 3975), 9-14 d (*Setifer* 2934, 2942, 3975, 8620, 8798, 9138), 9-15 d (*Tenrec* 2934, 2942; cf 4-6 d 6604), or 18 d (*Microgale* 2939).

Females have 6 (*Limnogale* 2934), 10 (*Setifer* 2934, 8798), or 21-29 mammae (*Tenrec* 6604, 7890, 9236). Males have abdominal testes (2662, 2934, 7890a, 8436) and the penis retracts into the cloaca (176). Details of endocrinology (7891) and birth (8622) are available.

## Order Insectivora

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| | | |
|---|---|---|
| **NEONATAL MASS** | | |
| *Solenodon paradoxus* | est 100 g | 2930 |
| **LITTER SIZE** | | |
| *Solenodon cubanus* | mode 1-2 | 10809 |
| *Solenodon paradoxus* | 1 emb; n1 f | 11853 |
| | 1 emb; n1 f | 176 |
| | 1 yg; n1 lt | 1285 |
| | mode 2; r1-3 | 7428 |
| | 3; n1 lt | 11269 |
| **CYCLE LENGTH** | | |
| *Solenodon paradoxus* | 9-13 d | 2930 |
| **GESTATION** | | |
| *Solenodon paradoxus* | est >50 d | 2930 |
| **LACTATION** | | |
| *Solenodon paradoxus* | wean: est 75 d | 2930 |

## Order Insectivora, Family Tenrecidae

The solitary, nocturnal tenrecs and their relatives (*Dasogale*, *Echinops* 110-250 g; *Geogale* 71 mm; *Hemicentetes* 70-170 g; *Leptogale*, *Limnogale* 50-90 g; *Microgale* 34-85 g; *Micropotamogale* 135 g; *Oryzorictes* 42-48 g; *Potamogale* 1 kg; *Setifer* 125-365 g; *Tenrec* 830-2400 g) are terrestrial, semiaquatic, or semifossorial predators on invertebrates in Madagascar and Africa. The reproductive biology of *Dasogale*, *Geogale*, *Leptogale*, *Limnogale*, and *Oryzorictes* is not known.

Male *Echinops* have an additional chromosome (3492). *Potamogale* has a simplex uterus with two nonfunctional cornu (4762), while other genera have bicornuate uteri (3218, 3855, 9236, 10484). Females have well-developed prostate and bulbourethral glands (6364, 8436). *Hemicentetes* (3218, 3857) and *Setifer* (8620) are reportedly polyestrous, although tenrecs may be induced ovulators (7984). Ovarian follicles lack antra, and fertilization occurs within the ovary (3218, 3219, 6169, 7892, 10484, 10485). Polyovular follicles are common (*Tenrec* 7890). *Hemicentetes* has no morula (3855), but *Tenrec* does (7892). The labyrinthine, hemochorial, discoidal or diffuse placenta has a circumplacental chorion, a hemophagous organ, but no consistent implantation site (1025, 3218, 3854, 3855, 3856, 4104, 4762, 4814, 7563, 7891, 10310, 10486). Placentophagia occurs (3975, 4949, 8620).

The naked neonates open their eyes at 7-9 d (*Echinops* 2942, 3844, 3975; cf 9-15 d 4949, 9138), 7-10 d (*Hemicentetes* 2934, 2942, 3454, 3975), 9-14 d (*Setifer* 2934, 2942, 3975, 8620, 8798, 9138), 9-15 d (*Tenrec* 2934, 2942; cf 4-6 d 6604), or 18 d (*Microgale* 2939).

Females have 6 (*Limnogale* 2934), 10 (*Setifer* 2934, 8798), or 21-29 mammae (*Tenrec* 6604, 7890, 9236). Males have abdominal testes (2662, 2934, 7890a, 8436) and the penis retracts into the cloaca (176). Details of endocrinology (7891) and birth (8622) are available.

| | | |
|---|---|---|
| **NEONATAL MASS** | | |
| *Echinops telfairi* | 3.8-9.6 g; smallest die | 3975 |
| | 7.7 g; n8 | 3844 |
| | 8.6 g; n9 | 3844 |
| *Hemicentetes semispinosus* | 11.5 g; r9.9-12.9; n4 yg, 1 lt | 3975 |
| *Microgale talazaci* | 3.6 g; n1, 1 d | 2939 |
| *Setifer setosus* | 24.7 g | 3975 |
| *Tenrec ecaudatus* | 10-18 g | 6604 |
| | r22.8-27.4 g; n4 | 2934 |
| | est 24 g | 7891 |
| **NEONATAL SIZE** | | |
| *Hemicentetes semispinosus* | 25 mm | 3454 |
| | 60-66 mm | 2934 |
| *Tenrec ecaudatus* | r84-92 mm; n4 | 2934 |
| **WEANING MASS** | | |
| *Tenrec ecaudatus* | 150-300 g; 5 wk | 6604 |
| **LITTER SIZE** | | |
| *Echinops telfairi* | r1-10 yg; n3 lt | 2942 |
| | 1.7; r1-3; n3 lt, 1 f | 8622 |
| | 3.4 yg; r2-5; n9 lt | 9138 |
| | 3.5; r2-12; n12 lt | 9523 |
| | mode 5 yg; r1-9; n25 lt | 3844 |
| | 5.4; r3-8; n5 lt | 4949 |
| | 5.85; r2-10; n13 lt, 12 f | 3975 |
| *Hemicentetes semispinosus* | 2; n1 lt | 3454 |
| | 3.6 yg; r1-11; n~36 lt | 2934, 2942, 3975 |
| | max 10 emb | 8436 |
| *Microgale dobsoni* | 2.67 yg; r2-3; n3 lt | 2939 |
| *Microgale talazaci* | r1-3; n4 lt | 2934, 2939 |
| *Micropotamogale ruwenzorii* | 2 emb; n1 f | 8852 |
| *Potamogale velox* | 1-2 yg | 4354a |
| | 1.4 emb, yg; r1-2; n5 lt | 2753 |
| *Setifer setosus* | 1-4 emb | 2942 |
| | 1-6 | 8620 |
| | 2.5 yg; r1-4; n2 lt | 2942 |
| | 2.6; r2-3; n5 lt | 5573 |
| | 2.6; r1-5; n8 lt | 9523 |
| | 2.7 yg; r1-5; n13 lt | 9138 |
| | max 6 emb | 10484 |
| | max 10 emb | 8436 |
| *Tenrec ecaudatus* | r1-4 yg; n3 lt | 2942 |
| | 10-32 emb | 2942 |
| | 10.5 cl; ±0.5; r3-18 | 7892 |
| | 12 emb; n1 f; 15 ovulations | 9236 |
| | 12.3 emb; r10-15; n3 f | 5573 |
| | 19.75 emb; r18-25; n4 f | 8906 |
| | 21 yg; n1 lt | 3854 |
| | 25 yg; r19-31; n2 lt | 6604 |
| | 24 emb; r16-32; n2 f | 3854 |
| | max 42 ovulations; max 32 emb | 1026 |
| **SEXUAL MATURITY** | | |
| *Echinops telfairi* | f: 1st conceive: 6 mo; n18 | 2942 |
| | f/m: 1 yr | 9523 |
| | f/m: 14 mo | 4949 |
| *Hemicentetes semispinosus* | f: 1st conceive: 30-35 d; n15 | 2934, 2942 |
| | f: 8 wk | 3975 |
| *Microgale dobsoni* | f/m: min 22 mo | 2939 |
| *Microgale talazaci* | f/m: ≥ 21 mo | 2939 |
| *Setifer setosus* | f: 1st conceive: 2 mo | 9138 |
| | f: 1st conceive: 6 mo; n2 | 2942 |
| | f: 1st birth: 14 mo; n1 | 8798 |
| | m: 8.5 mo; n1 | 8798 |
| *Tenrec ecaudatus* | f: 1st conceive: 6 mo; n1 | 2942 |

| | | |
|---|---|---|
| **CYCLE LENGTH** | | |
| *Echinops telfairi* | 6-7 d; r4-13; n7 cycles, 2 f | 3844 |
| **GESTATION** | | |
| *Echinops telfairi* | 49 d; n1 | 3844 |
| | est 7 wk | 9523 |
| | 60 d | 10797 |
| | 62-65 d; n5 | 2942 |
| | 66 d | 2934 |
| *Hemicentetes semispinosus* | ~4 wk; field data | 3218 |
| | ~45-55; n2 | 2942 |
| | >50 d; capture to birth | 3454 |
| | 55-58 d | 2934 |
| *Microgale dobsoni* | 58-64 d | 2930, 2939 |
| *Microgale talazaci* | 58-64 d | 2930, 2939 |
| *Setifer setosus* | ~5 wk; field data | 3218 |
| | ~7 wk; field data; not many | 5573 |
| | 51-56 d | 9138 |
| | 51-61 d; pairing to birth | 9523 |
| | 54-57 d; ibi or pairing to birth | 8620 |
| | 63 d | 2934 |
| | 65-69 d; n1 | 2942 |
| *Tenrec ecaudatus* | implant: 9.7 d; ±0.5 | 7892 |
| | 53 d | 6604 |
| | r58-64 d; n3 | 2934, 2942 |
| | est 60 d | 7891 |
| | ~9 wk; field data, not many | 5573 |
| **LACTATION** | | |
| *Echinops telfairi* | solid food: 12-14 d | 9138 |
| | solid food: 13 d | 3844 |
| | solid food: 16 d | 4949 |
| | wean: 30 d | 9138 |
| | independent: 30-35 d | 3844 |
| | wean: r18-22 d; n18 lt | 2942 |
| | wean: 23 d | 10797 |
| | wean: 32 d | 3975 |
| | wean: 33 d | 4949 |
| | wean: ~5 wk; field data | 3218 |
| *Hemicentetes semispinosus* | den emergence: 9 d | 2934 |
| | solid food: 13 d | 2934 |
| | wean: r18-25 d; n15 lt | 2942, 3975 |
| *Microgale dobsoni* | wean: 28-30 d | 2939 |
| *Microgale talazaci* | wean: 28-30 d | 2939 |
| *Setifer setosus* | solid food: 12-14 d | 9138 |
| | wean: 15-20 d; n2 lt | 2942 |
| | wean: 30 d | 9138 |
| *Tenrec ecaudatus* | nest emergence: est 15-20 d | 7891 |
| | solid food: 1 wk | 6604 |
| | wean: 25-30 d; n1 lt | 2942 |
| **LITTERS/YEAR** | | |
| *Potamogale velox* | 2 | 2753 |
| *Setifer setosus* | 2 | 5573 |
| *Tenrec ecaudatus* | 1; 2 rare | 2934 |
| **SEASONALITY** | | |
| *Echinops telfairi* | mate: est Nov-Dec; Madagascar | 3975 |
| | mate: all yr; captive | 4949 |
| | birth: Jan-Mar; Madagascar | 9523 |
| | birth: July-Aug; captive | 9523 |
| | birth: June-Sept, peak July; n15; zoo | 9138 |
| *Hemicentetes semispinosus* | mate: Sept-Dec; Madagascar | 2934 |
| | mate: Nov-May; Madagascar | 3975 |
| | birth: Nov-Jan; Madagascar | 2934 |
| | young found: yr rd; anecdotal reports; Madagascar | 8436 |
| *Microgale dobsoni* | mate: Dec-Aug; captive | 2934 |
| | birth: Feb-May; captive | 2934 |

| | | |
|---|---|---|
| *Potamogale velox* | preg f: Apr-May; few data; Cameroun | 9468 |
| | preg f: June; n2; s Cameroun, Benito, equatorial Guinea | 647 |
| | birth: early rainy season; Zaire | 11253 |
| *Setifer setosus* | mate: Sept, after Nov; ne Madagascar | 5573 |
| | mate: Sept-mid Oct; Madagascar | 2934 |
| | ovulate: Sept-Oct, no data other mo; Madagascar | 10484 |
| | preg f: Nov-Dec; n5; Madagascar | 5575 |
| | preg f/nearly full grown yg: Nov-Dec; ne Madagascar | 5573 |
| | birth: Jan-Feb, Apr-June, Aug-Nov, none Mar, July, Dec; zoo | 9138 |
| | birth: July, Dec, none Mar; few data?; captive | 8620 |
| | young found: yr rd; according to natives; Madagascar | 8436 |
| *Tenrec ecaudatus* | mate: Aug-Oct; Seychelles | 7892 |
| | mate: Oct-Nov; Madagascar | 2934 |
| | mate: annually; Seychelles | 7891 |
| | preg f: Dec-Jan; n3; Madagascar | 5573 |
| | preg f: Oct-Nov; n4; Madagascar | 8906 |
| | birth: Dec-Feb; Madagascar | 8436 |
| | birth: Jan-Feb; Madagascar | 2934 |
| | lact f: Feb; n1; Madagascar | 5573 |

## Order Insectivora, Family Chrysochloridae

Golden moles (*Amblysomus* 40-100 g, *Calcochloris* 20-30 g, *Chlorotalpa* 50-75 g, *Chrysochloris* 40-90 g, *Chrysospalax* 350-550 g, *Cryptochloris* est 20 g, *Eremitalpa* 15-20 g) are blind, subterranean carnivores from equatorial and southern Africa. Reproductive data for *Calcochloris*, *Chlorotalpa*, and *Cryptochloris* are lacking. Male *Amblysomus* are slightly larger than females (6136). The endothelial, discoidal, labyrinthine placenta has a superficial, mesometrial implantation into a bicornuate uterus (2394, 2662, 7563, 10310, 11108). Testes are abdominal and the penis is retractile (6136). A urogenital sinus is present (6136). Neonates are naked (6135, 6136, 10158) and suckle from 4 teats (6135, 10158).

| | | |
|---|---|---|
| **NEONATAL MASS** | | |
| *Amblysomus hottentotus* | ~4.5 g | 6136 |
| **NEONATAL SIZE** | | |
| *Amblysomus hottentotus* | HB: 47 mm | 6135 |
| *Chrysochloris stuhlmanni* | ~47 mm | 5759 |
| **LITTER SIZE** | | |
| *Amblysomus hottentotus* | 2 emb; n1 f | 10158 |
| *Chrysospalax villosus* | 2 emb; n1 f | 10158 |
| *Eremitalpa granti* | 1 emb ea; n2 f | 4927 |
| **LACTATION** | | |
| *Chrysochloris stuhlmanni* | nurse: 2-3 mo | 5759 |
| **SEASONALITY** | | |
| *Amblysomus hottentotus* | repro: yr rd; S Africa | 6136 |
| | birth: perhaps summer; S Africa | 10158 |
| *Chrysochloris asiatica* | birth: winter; S Africa | 10158 |
| *Eremitalpa granti* | mate: Oct-Nov; S Africa | 11108 |

## Order Insectivora, Family Erinaceidae

Gymnures and hedgehogs (*Echinosorex* 0.5-2 kg; *Erinaceus* 140-1400 g; *Hemiechinus* 220-400 g; *Hylomys* 15-80 g; *Neohylomys* 50-59 g; *Neotetracus* 101-111 mm; *Paraechinus* 150-450 g; *Podogymnura* 50-70 mm) are solitary, nocturnal, terrestrial insectivores in Eurasia and Africa. Male *Erinaceus* may be larger than females (2045; cf 4688), while the reverse is reported for *Echinosorex* (2313). Coitus may induce ovulation in *Erinaceus* (10416; cf 3795) and a vaginal plug is present after mating (*Erinaceus* 1570, 5759, 10416). The hemoendothelial, labyrinthine, discoid placenta has a decidica basalis and an eccentric, antimesometrial, interstitial implantation into a bicornuate or bipartite uterus with a massive cervix (2662, 6727, 7178, 7563, 8219, 9678, 9839, 10310). A postpartum estrus is not usual (2045, 2417, 6439, 6440, 7527, 7529, 7530, 12094).

Neonates are blind, but generally have quills (2951, 4158, 4690, 5224, 7226, 8049, 8718, 9175, 11438). Eyes open at 8-22 d in *Erinaceus* (69, 1301, 2045, 2384, 3571, 4687, 5759, 6440, 7529, 7530, 10153) and at 23-29 d in *Paraechinus* (2951). Females have 4-10 mammae (179, 2662, 2951, 6369, 7621, 10158). Testes are abdominal (2662, 6727). Males have seminal vesicles and females have homologous glands and a large clitoris (2662, 3984, 7454). A urogenital sinus is present (3985). Details of ovarian function (2417, 3436, 3795, 8219, 9839, 9842) and testicular function (165, 912, 1570, 1839, 2075, 2823, 2957, 3436, 3794, 3984, 3985, 9410, 9411, 9412, 9413, 9414, 10747) are available.

| | | |
|---|---|---|
| **NEONATAL MASS** | | |
| *Echinosorex gymnurus* | 14.5 g; r14-15; n2 | 7161 |
| *Erinaceus albiventris* | 6-6.5 g | 6440 |
| | 7-9 g | 1301 |
| | 8-11 g | 7226 |
| *Erinaceus concolor* | 22 g | 9632 |
| *Erinaceus europaeus* | 10.7 g | 69 |
| | 15 g; r8-22 | 7529, 7530 |
| | m: 16.1 g; r12.8-20.12; n7 | 4686, 4687, 4688 |
| | f: 17.17 g; r11.9-25.14; n5 | 4686, 4687, 4688 |
| *Erinaceus frontalis* | 9.8-11.3 g | 10154 |
| | m: 10.67 g; r9.5-12.0; n3 | 5224 |
| | f: 10.8 g; r10.5-11.5; n3 | 5224 |
| *Hemiechinus auritus* | 8 g; n1 | 9632 |
| | 8.325 g | 8718 |
| | 11 g; n1 | 11438 |
| | 13.15 g; n1 | 4690 |
| *Paraechinus aethiopicus* | 8-9 g; n3, 1 d | 2951 |
| **NEONATAL SIZE** | | |
| *Echinosorex gymnurus* | 80.5 mm; r80-81; n2 | 7161 |
| *Erinaceus europaeus* | 25 mm | 10338 |
| | 56.4 mm | 69 |
| | m: 68 mm | 4686 |
| | f: 91.5 mm; r87-96; n2 | 4686 |
| *Erinaceus frontalis* | f: HB: 52.3 mm; r51-53; n3 | 5224 |
| | m: HB: 52.5 mm; r52.5-53; n3 | 5224 |
| *Hemiechinus auritus* | 47 mm | 9175 |
| | 50 mm; n1 | 4690 |
| | HB: 61 mm; n1 | 11438 |
| **WEANING MASS** | | |
| *Erinaceus* spp | 306.8 g; ±57.4; n6 | 6042 |
| *Erinaceus europaeus* | 125-345 g; 40 d | 7529 |
| **WEANING SIZE** | | |
| *Erinaceus europaeus* | 160 mm | 2045 |
| **LITTER SIZE** | | |
| *Echinosorex gymnurus* | 1.9; r1-2; n7 lt | 7161 |
| | 2 emb; n1 f | 7178 |
| | 2 emb; n1 f | 557 |
| | 2 emb; n1 f | 2313 |
| *Erinaceus albiventris* | 2-4 yg | 4354a |
| | r2-3; n6 lt | 7226 |
| | 3 | 4693 |
| | 3.0; r1-6 | 1301 |
| | 4 emb; n1 f | 4924 |
| | 9; n1 lt | 6440 |
| *Erinaceus concolor* | 3 yg; n1 nest | 9632 |
| *Erinaceus europaeus* | 2 yg; n1 f | 2109 |
| | 2-3 cl | 3795 |
| | mode 3-8; r2-10 | 3571 |
| | 3.0; n113 lt | 7529, 7530 |
| | 3.86 yg; r3-5, n7 lt | 8922 |
| | 4-5; max 7; n67 lt | 6439 |
| | 4-7 emb | 2417 |
| | 4-8 | 69 |
| | 4.25 yg; r2-6; n76 nests | 7538 |

| | | |
|---|---|---|
| | 4.4 healthy emb, 4.6 implanted emb; n53 f | 7527 |
| | 4.5; r4-5; n2 f | 11255 |
| | 4.7; r3-7; n3 lt | 4686, 4687, 4688 |
| | 4.75; mode 4-6; r2-9 | 11399 |
| | 5 emb; n1 f | 1590 |
| | 5.2 wean; sd2.0; se0.2; n85 | 6035 |
| | 6.0 yg; r5-8; n3 f | 8049 |
| | 6.8 emb, yg; r5-8; n4 lt | 10338 |
| *Erinaceus frontalis* | 3-6 | 8965 |
| | 4; n1 lt | 4180 |
| | ~4; r1-9 | 10158 |
| | 5; r2-10 | 2384 |
| | 9; n1 lt | 5224 |
| *Hemiechinus auritus* | 1-6 | 9175 |
| | 2; n1 lt | 11438 |
| | 2.33; r1-5; n12 lt | 6727 |
| | 2.4 yg; r2-4; n5 lt | 8718 |
| | 2.5 yg; r2-3; n2 nests | 9632 |
| | 3 emb; n1 f | 7716 |
| | 4; n1 lt | 4690 |
| | 4 yg; n1 f | 8049 |
| | 4-7 | 4645 |
| | 5; r4-6; n3 lt | 8049 |
| | 5.8; r2-8 | 910 |
| | 6 yg; 1 lt | 4158 |
| *Hylomys suillus* | 2 emb; n1 f | 7161 |
| | 2-3 yg | 6369 |
| *Neotetracus sinensis* | 4.3 emb; r4-5; n3 f | 179 |
| *Paraechinus aethiopicus* | 2 emb; n1 f | 2951 |
| | 4 yg; n1 lt | 2951 |
| *Paraechinus hypomelas* | 3-4 | 9175 |
| | 4 yg; n1 f | 8049 |
| *Paraechinus micropus* | 1.5 yg; r1-2; n2 f | 8718 |
| | 4; n1 lt | 11438 |
| | 5 yg; 1 lt | 4158 |
| **SEXUAL MATURITY** | | |
| *Erinaceus albiventris* | f/m: 1st mate: 4 mo | 6440 |
| | f/m: 61-68 d | 1301 |
| | f/m: yr after birth | 5759 |
| *Erinaceus europaeus* | f: 1st birth: 11 mo | 7529, 7530 |
| | f: 1st birth: 1 yr | 4688 |
| | f/m: 5 mo; captive; 9 mo; wild | 6439 |
| | f/m: 10-12 mo | 3571 |
| **CYCLE LENGTH** | | |
| *Hemiechinus auritus* | 7.9 d; n16 cycles, 25 f | 9839 |
| | proestrus: 1.90 d; r1-3; metestrus: 1.8 d; r1-2.5 | 9839 |
| | estrus: 1.2 d; r0.5-2.2; diestrus: 2.5 d; 1.5-4.5 | 9839 |
| **GESTATION** | | |
| *Echinosorex gymnurus* | 35-40 d | 6369 |
| *Erinaceus albiventris* | 28-40 d; pairing to birth | 7226 |
| | 5-6 wk | 4693 |
| | mode 36 d; r34-44; pairing to birth | 1301 |
| | 37 d | 6440 |
| | r37-38 d; n3 | 4694 |
| *Erinaceus europaeus* | ~1 mo | 2662 |
| | 34-35 d; r31-39; n13 lt | 7527, 7529, 7530 |
| | 35-40 d | 8922 |
| | 5-6 wk; r34-49 d | 4687, 4688 |
| | 37 d; min ibi | 6439 |
| *Erinaceus frontalis* | 30-40 d | 2384 |
| | ~40 d | 10153 |
| *Hemiechinus auritus* | 30-32 d | 12094 |
| | 35-42 d | 9175 |
| | 36-37 d; n1 | 4690 |

| | | |
|---|---|---|
| | ~7 wk | 4645 |
| *Hylomys suillus* | 30-35 d | 6369 |
| *Paraechinus micropus* | est 35-39 d | 11438 |
| **LACTATION** | | |
| *Erinaceus* spp | solid food: 21-22 d | 6042 |
| | wean: 42.5 d; r42-43; n2 | 6042 |
| *Erinaceus albiventris* | solid food: ~1 mo | 5759 |
| | wean: ~40 d | 5759 |
| | solid food: 44 d | 8798 |
| | wean: 2 mo | 8798 |
| *Erinaceus europaeus* | den emergence: 3-4 wk | 7527, 7529, 7538 |
| | den emergence: 3-6 wk | 6035 |
| | solid food: 18-20 d | 3571 |
| | solid food: 23-24 d | 4687 |
| | solid food: 25-26 d; n2 lt | 7527, 7529 |
| | wean: 38-44 d | 7527, 7529, 7538 |
| | wean: 38-45 d; n5 lt | 8922 |
| | wean: 40+ d | 6439 |
| | wean: 45 d | 4687 |
| *Erinaceus frontalis* | forage: 6 wk | 10158 |
| *Hemiechinus auritus* | solid food: ~1 mo | 8049 |
| | den emergence: ~3 wk | 8049 |
| *Paraechinus aethiopicus* | solid food: 40 d | 2951 |
| *Paraechinus micropus* | wean: est ~5 wk | 11438 |
| **LITTERS/YEAR** | | |
| *Echinosorex gymnurus* | 2 | 6369 |
| *Erinaceus europaeus* | 1 | 11399 |
| | 1-2 | 2417 |
| | 1-2; 2 may not be exceptional | 10338 |
| | 1.5; r1-2; n2 | 4687 |
| | 2; n1 f | 7527 |
| *Hemiechinus auritus* | 1 | 9175 |
| | 1 | 4645 |
| *Paraechinus hypomelas* | 1 | 9175 |
| **SEASONALITY** | | |
| *Echinosorex gymnurus* | preg f: May-June, Sept, Nov; Madagascar | 7161 |
| | preg f: mid July-mid Aug; n1/5; N Borneo, Indonesia | 2313 |
| | repro: est yr rd; Thailand | 6369 |
| *Erinaceus albiventris* | estrus: yr rd; captive | 4693 |
| | mate: Apr; n1; captive | 6440 |
| | preg f: May; n1; Somalia, Ethiopia, Sudan | 4924 |
| | birth: May; n1; captive | 6440 |
| | birth: June, Sept, Oct; n3; captive | 4694 |
| *Erinaceus europaeus* | spermatogenesis: Jan-May; Europe | 2075 |
| | spermatogenesis: Apr-Aug, peak mid Apr-mid June; captive | 7527 |
| | spermatogenesis: May-Aug; central France | 3794 |
| | testes: large Mar-Apr, small Nov-Dec; France | 9413 |
| | testes: large Apr, small July-Jan; Europe | 2075 |
| | testes: large Apr-May; British Isles | 2662 |
| | testes: large Apr-Aug; Wales, England, UK | 165 |
| | testes: large May, small Sept, increase Jan; captive | 10747 |
| | testes: active Dec-Sept; France | 9413 |
| | cl present: Apr-Sept; Wales, England, UK | 2417 |
| | estrus: Apr-Sept; Germany | 9678 |
| | estrus: Oct, Mar; captive | 4693 |
| | mate: Mar; captive | 7530 |
| | mate: Mar-July; Switzerland | 850 |
| | mate: Mar-Aug; central Europe | 3571 |
| | mate: Apr-Sept; France | 3795 |
| | mate: end Apr-Aug/early Sept; captive | 7527 |
| | mate: end July-Sept; Denmark | 11399 |
| | mate: spring-summer; captive | 4687 |
| | preg f: May-June, Oct; British Isles | 2662 |
| | preg f: May-Sept; Great Britain | 7529 |

| | | |
|---|---|---|
| | birth: May-June, Sept-Oct; captive | 9410 |
| | birth: May/June-Sept; central Europe | 3571 |
| | birth: end May; captive | 7530 |
| | birth: end May-early June; 1 lt 23 June; USSR | 69 |
| | birth: June; n1; lab | 4686 |
| | birth: end July; captive | 4687 |
| | birth: peak Aug; Denmark | 11399 |
| | birth: Sept; n1; zoo | 2109 |
| | birth: mid Sept; captive | 7527 |
| | birth: end Sept; France | 3795 |
| | birth: spring; Britain, Ireland | 2045 |
| | birth: summer; central Europe | 4688 |
| | birth: winter; captive, Switzerland | 6439 |
| | repro: Apr-Aug; e Europe | 8798 |
| *Erinaceus frontalis* | mate: Nov; S Africa | 5224 |
| | birth: Oct-Mar; Botswana | 10154 |
| | birth: Oct-Mar/Apr; S Africa | 10158 |
| | birth: Nov; Zimbabwe | 10153 |
| *Hemiechinus auritus* | sperm: Mar-Aug; Rajasthan, India | 3985 |
| | testes: large Jan-Aug, small Sept-Dec; Jaipur, India | 402 |
| | testes: large summer; Agra, India | 6727 |
| | estrus: mid Feb-May; captive | 4693 |
| | estrus: Apr-Aug; Rajasthan, India | 9839 |
| | mate: after hibernation; Pakistan | 9175 |
| | birth: July-Sept; Rajasthan, India | 8720 |
| | birth: July; Stavropol, Russia | 910 |
| | birth: peak Aug-Sept; Pakistan | 9175 |
| | birth: early summer; Europe, w USSR | 4645 |
| | lact f: June; n1; e Europe, n Asia | 8049 |
| *Hylomys suillus* | preg f: Mar; n1; Malaysia | 7161 |
| | repro: est yr rd; Thailand | 6369 |
| *Paraechinus aethiopicus* | lact f: June; n1; Egypt | 11468 |
| *Paraechinus hypomelas* | birth: end spring-early summer; Pakistan | 9175 |
| | lact f: May; n1; s Turkmenia | 8049 |
| *Paraechinus micropus* | repro: seasonal; India | 912 |

## Order Insectivora, Family Soricidae

Shrews (*Anourosorex* 14-35 g; *Blarina* 7-30 g; *Blarinella*, *Chimarrogale* 28-32 g; *Crocidura* 2-66 g; *Cryptotis* 3-9 g; *Diplomesodon* 7-13 g; *Feroculus* 35 g; *Megasorex* 9.5-11.7 g; *Myosorex* 7-28 g; *Nectogale*, *Neomys* 6-23 g; *Notiosorex* 3-6 g; *Paracrocidura* 6-16 g; *Scutisorex* 70-113 g; *Solisorex* 59-66 mm; *Sorex* 2-25 g; *Soriculus* 4-14 g; *Suncus* 1-110 g; *Sylvisorex* 3-92 g) are terrestrial to semiaquatic, solitary insectivores from Eurasia, Africa, North and Central America, and northern South America. Males have been reported to be larger than females (*Blarina* 8299, *Crocidura* 4601, *Myosorex* 9329; cf 673, *Sorex* 716, 5789, 9159; cf 1912, 3423, 7893, and *Suncus* 572, 1331, 8501, 8903, 11999a).

Ovulation may be induced by coitus (1210, 1217, 2001, 2737, 6728, 7410, 7420, 7524, 8299, 8757, 10677). A copulatory tie is reported in *Blarina* (8299), *Cryptotis* (7409, 7410), and perhaps *Neomys* (7273), but is not present in *Sorex* (2155) or *Suncus* (3374). A vaginal plug is reported for *Neomys*, as is intrauterine insemination (3630, 8757). Conception occurs within the follicle and corpora lutea development begins before ovulation (2737, 8299, 10482).

The endothelio-endothelio, endotheliochorial, or hemochorial, labyrinthine, villous, discoid placenta has an antimesometrial, superficial implantation into a bipartite or bicornuate uterus (231, 373, 1210, 1221, 2737, 4480, 5734, 5823, 7410, 7419, 7563, 7573, 8171, 8299, 9481, 9483, 9840, 9841, 10019, 10310, 11814) and may be eaten after birth (335, 5118). A postpartum estrus occurs (1210, 1217, 2004, 2045, 2155, 2156, 2737, 3373, 4600, 4601, 4604, 4894, 5244, 5373, 7273, 7409, 7410, 8757, 9175, 9916, 10017, 10019, 10482, 11313, 11315, 11316; cf 10339) and may be accompanied by delayed implantation (1210, 5039, 8058a).

Neonates are blind and naked (187, 1013, 1474, 2347, 2437, 3375, 3455, 3571, 4266, 4279, 4901, 5146, 7161, 7739, 8049, 9175; cf 335, 2589, 2736, 9914, 11314). Their eyes open at 5 d (*Blarina* 7333), 9-13 d (*Crocidura* 335, 2589, 3458, 3571, 4602, 4603, 10158, 11313, 11314, 11391; cf 11473), 1-2 wk (*Cryptotis* 2004,

7410; *Suncus* 2736, 2744, 3373, 9175, 9914, 11315), 15-18 d (*Myosorex* 673), 15-21 d (*Sorex* 2045, 2156, 3423, 3571, 5118, 5373, 9175, 11314, 11390; cf 1474), 20-24 d (*Neomys* 2156, 7272, 7275, 7276).

Females have 4-6 mammae (*Sylvisorex* 7260), 6 mammae (*Blarina* 1474, 4266; *Crocidura* 331, 2575, 3571, 4602, 4924, 5405a, 7167a, 7621, 8049, 9945; cf 8798; *Cryptotis* 4494; *Sorex* 502, 1474, 2548, 7621, 9175), or 10-14 mammae (*Neomys* 7274, 8757). The vagina may be sigmoid during estrus (*Blarina* 3725).

Males have a sigmoid penis which may have grooves or spicules, abdominal testes which may migrate seasonally (*Blarina* is ascrotal), and seminal vesicles which may not be present in all genera (373, 1211, 2742, 2845, 3984, 4277, 6888, 6945, 7438, 7441, 8299, 8774). *Sorex araneus* males have an additional chromosome (3492).

Details of anatomy (1210, 4827, 4828, 6019, 8757, 10558, 10676, 11677, 11678), intersex (1216), endocrinology (2743, 3565, 4605, 11807; *Suncus* 494, 2738, 2739, 3161, 4476, 4480, 4481, 6728, 7170, 9146, 9147, 9149), spermatogenesis (2742, 3603, 3604, 9995), sexual behavior (1900, 9143, 9144, 9145, 9146, 9148, 9153), milk composition (2740, 7581), photoperiod (9151, 11499), and reproductive energetics (3698, 7581, 7582) are available.

**NEONATAL MASS**

| | | |
|---|---|---|
| *Blarina brevicauda* | 1.34 g; n1, 36-48 h | 4266 |
| *Crocidura bicolor* | 0.25 g ea; n2 | 11473 |
| | 1.4 g; n1, 1 d | 2589 |
| *Crocidura flavescens* | 2.26 g; sd0.20; n2 | 3698 |
| *Crocidura hirta* | ~1 g | 7167a |
| *Crocidura leucodon* | 0.8 g | 3455 |
| | 1.0 g; n2 | 3458 |
| | 1 g | 3571 |
| *Crocidura russula* | 0.78 g; r0.5-1.1; n39 | 4602, 4603 |
| | 1.11 g; sd0.23; n40 | 3698 |
| *Crocidura suaveolens* | 0.53 g; r0.42-0.67 | 11313, 11314 |
| *Crocidura viana* | 1.39 g; sd0.23; n13 | 3698 |
| *Cryptotis parva* | 0.3-0.4 g | 7409 |
| | 0.32 g | 4279 |
| *Myosorex varius* | ~1 g | 673 |
| *Neomys anomalus* | 0.52-0.61 g | 7272 |
| *Neomys fodiens* | 0.53-0.89 g; n72 yg, 9 lt | 7275 |
| | 0.62 g; n34 yg, 3 lt before nursing | 7275 |
| | 1.0 g | 2156 |
| *Sorex araneus* | 0.391 g; r0.345-0.431; n16 | 11314 |
| | 0.4-0.5 g | 3571 |
| | 0.42 g | 1838 |
| | 0.5 g | 2156 |
| *Sorex cinereus* | ~0.1 g | 5146 |
| | 0.28 g; se0.02; r0.23-0.36; n15 | 3423 |
| *Sorex coronatus* | 0.57 g; sd0.10; n12 | 3698 |
| *Sorex minutus* | 0.22 g; n1 | 3698 |
| | 0.25 g | 2156 |
| *Sorex sinuosus* | 0.5 g | 7869 |
| *Sorex vagrans* | ~0.5 g | 5373 |
| *Suncus etruscus* | 0.18-0.27 | 3373 |
| | 0.19 g; r0.17-0.21; n2 | 11315 |
| | 0.20 g | 3374, 3375 |
| *Suncus murinus* | f: 2.0 g; r1.4-3.4; n32 some with 0.2 g milk | 2736, 2744 |
| | 2.1 g; se0.1; r1-4; n211 lt | 4477 |
| | f: 2.2 g; sd0.4; n7 | 9914 |
| | m: 2.3 g; sd0.3; n6 | 9914 |
| | m: 2.3 g; r1.6-3.1; n28 some with 0.2 g milk | 2736, 2744 |
| | 2.7 g; n40 lt, 82 yg | 2741 |
| | 2.8 g; se0.2; r1-6; n51 lt | 4477 |
| | 3.1 g; n1 lt, 4 yg | 7161 |
| | m: 3.6 g; sd0.6; n4 | 9914 |
| | f: 3.7 g; sd0.4; n2 | 9914 |

| | | |
|---|---|---|
| **NEONATAL SIZE** | | |
| *Blarina brevicauda* | TL: 31 mm; 2 d | 3725 |
| *Crocidura bicolor* | HB: 18.7 mm ea; n2 | 11473 |
| | HB: 20 mm; 1 d | 2589 |
| | HB: ~20 mm; n3 | 335 |
| *Crocidura hirta* | HB: ~30 mm | 7167a |
| *Crocidura russula* | HB: 29 mm; n311 | 4603 |
| *Myosorex varius* | HB: 30 mm | 673 |
| *Neomys fodiens* | 20.5 mm; r17.0-22.5; n8 | 7275 |
| *Sorex araneus* | 18 mm | 1838 |
| *Sorex cinereus* | CR: 12-14 mm | 3423 |
| *Sorex minutus* | 16 mm | 5118 |
| *Suncus etruscus* | HB: 14.2-15.4 mm | 3373, 3375 |
| **WEANING MASS** | | |
| *Crocidura bicolor* | 3.0 or 3.4 g; 23 d | 2589 |
| *Crocidura flavescens* | 20.7 g; sd1.4; n3 | 3698 |
| *Crocidura hirta* | ~10 g; 15-20 d, from graph | 7167a |
| *Crocidura russula* | f: 6.8 g; r5.01-9.0; n49 | 4603 |
| | m: 7.3 g; r4-10; n52 | 4603 |
| | 9.0 g; sd0.7; n19 | 3698 |
| *Crocidura suaveolens* | ≤4 g | 9253 |
| *Crocidura viana* | 13.7 g; sd1.9; n13 | 3698 |
| *Cryptotis parva* | at adult size | 7409 |
| *Neomys fodiens* | 10 g | 2156 |
| *Sorex araneus* | ~7 g; 18 d | 2045 |
| | 10 g | 2156 |
| *Sorex cinereus* | 3.5 g; ±0.02; r3.5-3.6; n6 | 3423 |
| *Sorex coronatus* | 8.6 g; sd0.9; n9 | 3698 |
| *Sorex fumeus* | 4 g; ~1 mo | 4277 |
| *Sorex minutus* | 3.2 g; n1 | 3698 |
| | 7 g | 2156 |
| *Sorex vagrans* | 5-6 g; ~1 mo | 5373 |
| *Suncus etruscus* | ~2 g | 11315 |
| | 2.05-2.21 g; 19-21 d | 3373 |
| *Suncus murinus* | f: 13.2 g; n19, 20 d | 2736 |
| | m: 15.0 g; n19, 20 d | 2736 |
| | f: 28.2 g; sd2.3; n9 | 9914 |
| | m: 34.2 g; sd5.5; n11 | 9914 |
| | f: 46.6 g; sd8.2; n14 | 9914 |
| | m: 62.9 g; sd15.7; n11 | 9914 |
| **WEANING SIZE** | | |
| *Crocidura russula* | 66.1 mm | 4603 |
| **LITTER SIZE** | | |
| *Anourosorex squamipes* | 3.2 emb; r2-4; n4 f | 148 |
| *Blarina brevicauda* | 4 emb; n1 f | 2547 |
| | 4 emb; n1 f | 5798 |
| | 4.5 emb; r4-5; n2 f | 4280 |
| | 4.7 bom; se0.2; r1-8; 3.41 weaned; n81 lt | 1028 |
| | 4.75 emb; r4-6 n2 f | 5405b, |
| | 5 emb; n1 f | 495 |
| | 5; n1 lt | 7333 |
| | 5 yg; n1 | 6882 |
| | 5 emb; n1 f | 7235 |
| | 5.3 emb; r2-8; n32 f | 8299 |
| | 6-7 emb, yg; r5-10; n29 f | 4266 |
| | mode 6; r4-8 | 9164 |
| | 6.1 cl; n23 f | 8299 |
| | 6.3 emb; r5-8; n8 f | 1426 |
| | 6.6 ps, emb; r5-8 | 10839 |
| | 6.8; r5-8; n6 lt | 11956 |
| *Blarina carolinensis* | 3 emb; n1 f | 6298 |
| | 3.7 emb; r3-5; n3 f | 1290 |
| | 4 emb; n1 f | 7498 |
| *Crocidura allex* | 4 emb; n1 f | 4924 |

| | | |
|---|---|---|
| *Crocidura attenuata* | 1 emb; n1 f | 179 |
| *Crocidura bicolor* | 2; n1 lt | 11473 |
| | 2.0; r1-3; n2 lt | 4924 |
| | 3; n1 lt | 335 |
| | 3 yg; n1 lt | 2589 |
| | 4 emb; n1 f | 7167a |
| | 4 emb; n1 f | 10154 |
| *Crocidura bottegi* | 3 emb; n1 f | 4354a |
| *Crocidura buttikoferi* | 3 emb; n1 f | 2955 |
| *Crocidura crossei* | 2-4 yg | 4354a |
| *Crocidura cyanea* | 2 emb; n1 f | 10154 |
| | 3.6 emb; r2-6; n5 f | 10158 |
| | 4.7 emb; r4-6; n3 f | 7167a |
| *Crocidura dracula* | 2 emb; n3 f | 179 |
| *Crocidura dsinezumi* | 4.3 emb; r3-5; n3 f | 5405a |
| *Crocidura eisentrauti* | 2 emb; n1 f | 2955 |
| *Crocidura flavescens* | 2 emb; n1 f | 1948 |
| | 2 yg; n1 f | 6828a |
| | 2.28 emb; mode 2; r1-4; n29 f | 2575 |
| | 3.0 emb; mode 3; r2-4; n6 f | 2444 |
| | 3.8 yg; sd1.5; n5 lt | 3698 |
| | mode 4 yg | 4354a |
| | 4 emb; n1 f | 4924 |
| | 4 emb; n1 f | 6828a |
| | 4 emb; n1 f | 9945 |
| | 4.2; r3-5; n12 lt | 9893 |
| *Crocidura foxi* | 2 emb; n1 f | 4354a |
| *Crocidura fuliginosa* | r1-2; n3 lt | 7161 |
| *Crocidura fumosa* | 3.75 emb; r1-6; n4 f | 4924 |
| *Crocidura gracilipes* | 2 emb; n1 f | 2575 |
| | 3 emb; n1 f | 2443 |
| | 3 emb; n1 f | 2444 |
| | 3.25 emb; r3-4; n4 f | 4924 |
| *Crocidura hirta* | 2; r1-2; n3 lt | 187 |
| | 2.5 yg; r1-4; n4 lt | 7167a |
| | 3.4; r1-9; n9 | 10158 |
| | 3.6; r3-4; n5 | 10158 |
| | 3.8; r2-5; n7 lt | 8965 |
| | mode 4; r1-5 | 2449 |
| | 4 emb; n1 f | 6272 |
| | 4.0 emb; r3-5; n2 f | 9945 |
| | 4.2 emb; r4-5; n5 f | 7167a |
| | 4.25 emb; r2-6; n4 f | 331 |
| | 4.3; r2-6; n13 lt | 9892, 9893 |
| *Crocidura horsfieldi* | 3 emb, yg ea; n3 lt | 8501 |
| | 3.3 emb; r3-4; n3 f | 7382 |
| *Crocidura jacksoni* | 3.5 emb; r3-4; n2 f | 4924 |
| *Crocidura kivuana* | 1.67 emb; r1-2; n3 f | 2575 |
| *Crocidura lanosa* | 3.3 emb; r2-6; n3 f | 2575 |
| *Crocidura leucodon* | 2; n1 lt | 3458 |
| | 3; n1 lt | 3455 |
| | mode 3-6 | 3571 |
| | 3.37 yg; n19 lt | 4602 |
| | 5-10 | 8798 |
| *Crocidura littoralis* | 3.0 emb; r2-4; n2 f | 2575 |
| *Crocidura luna* | 2 emb; n1 f | 331 |
| *Crocidura mariquensis* | 3.3 emb; r2-5; n22 f | 10154 |
| | 3.7 emb; r2-5; n14 f | 9892, 9893 |
| | 4.2 emb; r2-7 | 8965 |
| | 4.5 emb; r4-5; n2 f | 7167a |
| *Crocidura nigricans* | 4 emb; n1 f | 9893 |
| *Crocidura niobe* | 3 emb; n1 f | 2575 |
| *Crocidura planiceps* | 3 yg; n1 nest | 4354a |

| | | |
|---|---|---|
| *Crocidura poensis* | 2 emb; n1 f | 1948 |
| | 3 emb; n1 f | 2955 |
| *Crocidura russula* | 2-6 | 5485 |
| | 3-10 | 3571 |
| | 3.0 yg; mode 4; r1-7; n937 lt | 4600, 4601, 4602 |
| | 3.5; r1-7; n86 lt | 11316 |
| | 3.65 emb | 6574 |
| | 4.1; r3-6; n9 lt | 955 |
| | 4.4 yg; sd1.3; n49 lt | 3698 |
| | 4.5; r4-5; n2 lt | 7898 |
| | 5-10 | 8049 |
| *Crocidura silacea?* | 4 emb; n1 f | 334 |
| *Crocidura suaveolens* | 2 emb; n1 f | 5096 |
| | 2.97 emb; r1-5; n31 f | 9253 |
| | 3.93; r2-5 | 11313 |
| | 5-10 | 8049 |
| *Crocidura turba* | 2; n1 lt | 2372 |
| *Crocidura usambarae* | 2 emb | 5032 |
| *Crocidura viaria* | 3.5 yg; sd1.0; n24 lt | 3698 |
| *Crocidura zaodon* | 2.5 emb; r2-3; n2 f | 4924 |
| *Crocidura zarudnyi* | 6 emb; n1 f | 2404a |
| *Cryptotis magna* | 3 emb ea; n2 f | 9177 |
| *Cryptotis mexicana* | 3 emb; n1 f | 4228 |
| *Cryptotis parva* | 2.67 emb; r2-3; n3 f | 6303 |
| | 3 yg; n1 nest | 1924a |
| | 3.33 emb, yg; r3-4; n3 f | 2347 |
| | 4 emb; r3-5; n3 f | 1290 |
| | 4.3 yg; r1-9 | 7409 |
| | 4.56 | 7410 |
| | 4.9 emb; r2-7 | 11626 |
| | 5-10 | 8798 |
| | 5 emb; n1 f | 1812a |
| | 5.0 yg; r4-6; n5 f | 2004 |
| | 6 yg; n1 nest | 11609 |
| | 8 emb; n1 | 6298 |
| *Diplomesodon pulchellum* | 5 yg | 4645 |
| *Myosorex babaulti* | 3 or 4 emb; n1 f | 2575 |
| *Myosorex cafer* | 3.0 emb; r2-4; n3 f | 10158 |
| | 3 emb; n1 f | 2096 |
| *Myosorex geata* | 4 yg | 5759 |
| *Myosorex norae* | 1; r1-2 | 5759 |
| | 2 emb; n1 f | 4924 |
| *Myosorex polulus* | 1-2 | 5759 |
| | 1.5 emb; r1-2; n2 f | 4924 |
| *Myosorex varius* | 2.8 emb; r1-4; n19 f | 9329 |
| | 2.9; mode 3; n10 lt | 673 |
| | 3 yg; n1 f | 3977 |
| *Neomys anomalus* | mode 5-6; r3-11 | 3571 |
| | 4.0; r2-5; n4 lt | 7272 |
| | 13 emb; n1 f | 5998 |
| *Neomys fodiens* | 3; n1 lt | 7898 |
| | 3-8 | 1123 |
| | 3.5; r2-5; n2 lt | 3630 |
| | 6.1 yg; max 10 | 11316 |
| | 6.28 yg; sd2.78; mode 6; r3-15; n18 lt | 7273 |
| | 6.38 yg; n29 lt | 7275 |
| | 6.7 yg; r3-15 | 7276 |
| | 6.8 emb; r3-8 | 8757 |
| | 8.0 emb; r5-11; n2 f | 1590 |
| | 11 emb; n1 f | 11508 |
| | 11 emb; n1 f | 6438 |
| *Notiosorex crawfordi* | 3 emb; n1 f | 8754 |
| | 3 emb; n1 f | 2617 |
| | 3 yg; n1 nest | 8754 |

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| | 3.5 emb, yg; r3-5; n4 f | 4901 |
| | 5 emb; n1 f | 529 |
| | 5 emb; n1 f | 1871b |
| | 5 emb; n1 f | 5410 |
| *Scutisorex somereni* | 1.86 emb; mode 2; r1-3; n14 f | 2575 |
| *Sorex alpinus* | 6 | 3571 |
| *Sorex araneus* | 1-11 emb | 10677 |
| | 2-8 | 1123 |
| | 3-8 emb | 9771 |
| | 4; r1-6 | 9916 |
| | 4.33 yg; r2-7; n6 lt | 3190 |
| | 4.5 emb; r2-8; n4 f | 3190 |
| | 4.8-5.4 | 9712 |
| | mode 5-7; r2-10 | 3571 |
| | 5-10 emb | 10482 |
| | 5 emb; n1 | 1590 |
| | 5.14; n29 lt | 9712 |
| | 5.25 cl; r3-8; n4 f | 3190 |
| | 5.3 emb; r4-7; n3 f | 7901 |
| | 5.40; r4-11; n30 young f | 10346 |
| | 5.46 emb; r1-9 | 1210 |
| | 5.8; r4-10; n8 lt | 11314 |
| | 5.9; r2-8; n12 lt | 11316 |
| | 5.92 yg; r2-11; n72 lt | 10677 |
| | 6-7 cl | 10482 |
| | 6; n1 lt | 11390 |
| | 6.0 emb; r4-8; n6 yg f | 8773 |
| | 6.45 emb; r1-9; n51 f | 1210 |
| | 6.91; r2-8; n82 adult f | 10346 |
| | 7.1 emb; r4-11; n66 f | 4577 |
| | 7.3 emb; r4-9; n9 f | 1838 |
| | 7.3; r5-9; n9 lt | 10339 |
| | 7.35 cl; mode 7; r4-12; n97 f | 1210 |
| | 7.5 emb; n15 f | 8372 |
| | 7.5-8.3; mode 7-9; r4-11; across populations | 10017 |
| | 7.7; mode 7; r1-10 | 2156 |
| | 7.75 emb; mode 8; n4 f | 5039 |
| | 7.8; r6-10 | 9159 |
| | 7.9 emb; r6-10; n8 f | 10344 |
| *Sorex arcticus* | 6.0 emb; r5-7; n2 f | 10144 |
| | 6.0 emb; r4-8; n5 f | 1912 |
| | 6.0 emb; r5-8; n12 f | 1426 |
| | 6.25 emb; r3-9; n4 f | 506 |
| | 7.25 emb; r6-9; n4 f | 5210 |
| | 7.3 cl; se0.36; r5-9; n15 f | 506 |
| | 7.5 emb; r6-9; n2 f | 495 |
| | 7.7 emb; se0.42; r5-9; n10 f | 506 |
| | 10 emb; n1 f | 716 |
| *Sorex caecutiens* | 4.0 cl; r3-5; n2 f | 8058a |
| | 5.9-8.9 | 8058a |
| | 7.0 emb; r5-9; n3 f | 8058a |
| | 7.1 emb; n8 f | 8058a |
| *Sorex cinereus* | 2-10 | 1474 |
| | 3 emb; n1 f | 574 |
| | 4-8 yg | 502 |
| | 4 emb; n1 f | 6166 |
| | 4.6 yg; r3-12 | 1871a |
| | 6.0 emb; r5-7; n2 f | 495 |
| | 6.5 emb; r6-7; n2 f | 12049 |
| | 6.5 emb; r6-7; n2 f | 10839 |
| | 6.7; mode 7; r5-9 | 3423 |
| | 6.8 emb; r5-7; n23 f | 1426 |
| | 7 emb; n1 f | 8808 |
| | 7 yg; n1 nest | 8766 |

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| | 7.91 emb; se0.4; r3-12; n32 f | 5147a |
| | 8 emb; n1 f | 2545 |
| | 8 emb; n1 f | 11956 |
| | 9.0 emb; r7-12; n3 f | 6553 |
| *Sorex coronatus* | 3.692 emb; ±1.109; r2-6; n13 | 6573 |
| | 5.1 yg; sd1.5; n19 lt | 3698 |
| *Sorex cylindricauda* | 5 emb; n1 f | 7382 |
| *Sorex daphaenodon* | 6 emb; n1 f | 12105 |
| *Sorex dispar* | 2 emb; n1 f | 5799 |
| | 2 emb; n1 f | 10683a |
| | 5 emb ea; n2 f | 9111a |
| *Sorex fontinalis* | 5; r4-6 | 9164 |
| *Sorex fumeus* | 5.1 emb | 8155 |
| | 5.5 emb; r2-8; n42 f | 4277 |
| | 6.6; r6-8 | 11956 |
| | 9 emb; n1 f | 10144 |
| *Sorex gracillimus* | 3.5 cl; r3-4; n2 f | 8058a |
| | 6.0 live emb ea; n2 f (1 f with an additional resorbing emb) | 8058a |
| *Sorex haydeni* | 4-10 yg | 1871a |
| *Sorex hoyi* | 5.3 emb; r3-8; n3 f | 6554 |
| | 6 emb; n1 f | 4525 |
| | 7 emb; n1 f | 9700 |
| *Sorex isodon* | 7.1 emb; se0.22; mode 7; r1-10; n61 f | 10019 |
| *Sorex longirostris* | 3.9 emb; sd1.36; r1-6; n15 f | 3512 |
| | 4 emb; n1 f | 2821b |
| | mode 4 | 1474 |
| | 6-10 yg | 3511 |
| *Sorex merriami* | 6.0 emb, swellings; r5-7; n3 f | 5352 |
| *Sorex minutus* | 2 emb; n1 f | 9771 |
| | mode 5-6; r2-9 | 1123 |
| | 5; n1 lt | 7901 |
| | 5.57 cl; mode 6; max 7; n21 f | 4001 |
| | 6 emb; n1 f | 7382 |
| | 6.0; r4-9; n9 f | 8773 |
| | 6.0 emb; r2-10; n3 f | 9159 |
| | 6.23 emb; r2-8; n31 f | 1217 |
| | 6.5 emb; r6-7; n2 f | 10339 |
| | 6.77 cl; r5-8; n39 f | 1217 |
| | 9 yg; n1 lt | 3698 |
| *Sorex monticolus* | 4-7 | 1474 |
| | 7 emb; r6-8; n2 f | 1871a |
| *Sorex nanus* | 6-7 yg | 1871a |
| | 6.5 emb; r6-8; n4 | 4894 |
| *Sorex ornatus* | 4-6 emb | 8156 |
| *Sorex pacificus* | 4.2 emb; se0.2; r3-5; n20 f | 1613, 1614 |
| | 4.8 ps; se0.5; r3-6; n6 f | 1614 |
| *Sorex palustris* | 3 emb | 2001 |
| | 4.7; r3-6; n3 f | 7809 |
| | 5 emb; n1 f | 6560 |
| | 5.43 emb, ps; r1-8; n14 f | 2001 |
| | mode 6 emb; r3-10 | 760 |
| | 6 emb; n1 f | 1119 |
| | 8 emb; n1 f | 6553 |
| *Sorex raddei* | 2.7 emb; n3 f | 10274 |
| *Sorex trowbridgii* | 3.77 ps; mode 4; r3-4; n13 f | 3623 |
| | 3.89 emb; mode 4; r3-5; n28 f | 3623 |
| | 3.94 cl; mode 4; n34 f | 3623 |
| | 5; r3-6; n8 f | 5244 |
| *Sorex tundrensis* | 12 emb; n1 f | 8808 |
| *Sorex unguiculatus* | 5.0 cl; sd1.5; r2-8; n12 non-preg f | 8058a |
| | 5.5 emb >8 mm; sd0.5; r5-6; n11 f | 8058a |
| | 5.6 emb <8 mm; sd1.0; r4-8; n20 f | 8058a |
| *Sorex vagrans* | 5 emb; n1 f | 6488 |
| | 5.16 emb, yg; r2-9; n39 f; 5.55 emb; 4.74 yg | 5373 |

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| | 5.6 emb; se0.43; r4-7; n7 f | 11230 |
| | 6.4 emb; r2-9; n32 f | 1910 |
| | 6.5 emb; r5-8; n2 f | 6553 |
| | 9 emb; n1 f | 4896 |
| *Soriculus caudatus* | 4.75 emb; r3-6; n8 f | 7382 |
| *Soriculus nigrescens* | 4.89 emb; r3-7; n9 f | 7382 |
| *Suncus etruscus* | 3.76 yg; mode 4; r2-5; n45 lt, 14 f | 3373, 3374 |
| | 4.0; r2-5; n5 lt | 11315 |
| | 4-6 | 9175 |
| | 5 emb; n1 f | 7923 |
| *Suncus lixus* | 3 emb; n1 f | 328a, 331 |
| *Suncus murinus* | 1-4 cl | 2737 |
| | 1.84; r1-3; n45 lt | 2736, 2740, 2741, 2744 |
| | 2 emb ea; n3 f | 8501 |
| | 2.7 emb; se0.14 | 572 |
| | 2.7; mode 3; r1-5; n71 lt | 4406 |
| | 2.7; sd1.3; mode 2-3; r1-5; n30 | 9914 |
| | 2.99 emb; sd1.2; r1-7 | 1331 |
| | 3.0; sd1.2; mode 2-3; r1-5; n23 | 9914 |
| | 3.21 yg; mode 3; n24 lt | 7524 |
| | 3.5 emb; n2 f | 9265 |
| | 3.5; ±0.16 | 11999a |
| | 3.67 emb; mode 3; n67 f | 7524 |
| | 3.8 emb; sd0.286; est n44 f | 6589 |
| | 3.85 emb; r1-8; n7 f | 7382 |
| | 4-5 yg | 8410a |
| | 4.71 emb; r3-6; n21 f | 8903 |
| *Suncus varilla* | 3 emb; n1 f | 8965 |
| | 3.3 yg; ±1.2; r2-7; n16 lt | 6658 |
| | 3.7 emb; ±1; r2-6; n30 f | 6658 |
| | 4 emb; n1 f | 6272 |
| *Sylvisorex granti* | 1.6 emb; r1-2; n5 f | 2575 |
| | 2; n1 lt | 187 |
| *Sylvisorex lunaris* | 2.7 emb; r2-4; n3 f | 2575 |
| *Sylvisorex megalura* | 1.8 emb; mode 2; r1-2; n5 f | 2575 |
| | 2 emb; n1 f | 330, 331 |
| *Sylvisorex morio* | 2 emb; n3 f | 2955 |
| **SEXUAL MATURITY** | | |
| *Blarina brevicauda* | f: receptive: 47 d; n1 | 8299 |
| | f: 1st conceive: 1-2 mo | 2261 |
| | f: 1st birth: 64 d; n1 | 1028 |
| | m: 1st sperm 1-2 mo | 2261 |
| | m: 1st mate: 85 d; n1 | 8299 |
| | f/m: fall born: 5 mo; spring born: <5 mo | 8299 |
| | f/m: mate: 10 wk | 7333 |
| *Crocidura leucodon* | f/m: 3 mo | 8798 |
| | f/m: 4 mo | 3571 |
| *Crocidura russula* | f: 1st mate: 58-416 d; n13; 9/13 <94 d | 11316 |
| | f: 1st mate: 7-15 wk | 4600, 4601, 4602, 4604 |
| | f: 1st birth: 13.6 wk; r11-19; n15 | 4601 |
| | m: 1st sperm: 30-32 d | 4600, 4601 |
| | m: 1st mate: 71-642 d; n9; 4/9 <114 d | 11316 |
| | f/m: 1-3 mo | 5276a |
| *Crocidura suaveolens* | f: 1st mate: 2 mo; n1 f | 9253 |
| | f/m: 1st mate: 1 yr | 11313 |
| *Cryptotis parva* | f: 1st mate: 31 d | 7410 |
| | f: 31 d | 7409 |
| | m: 1st sire: 36 d | 7409, 7410 |
| *Neomys anomalus* | f/m: 1st mate: 3-4 mo | 3571 |
| *Neomys fodiens* | f/m: 1st mate: rarely yr of birth; most 2nd season | 8757 |
| | f/m: 1st mate: 3-4 mo | 3571 |
| *Sorex araneus* | f: 3-5 mo; n "a few" | 3571 |
| | f: 4.6% 4-6 mo; mode 10-11 mo | 4577 |
| | f: ~9 mo-1 yr | 10339 |
| | m: testes large: est yg of yr; n1; 6 g | 5982 |

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|---|---|---|
| | m: testes large: 10-11 mo | 4577 |
| | m: testicular sperm: ~1 yr | 1211 |
| | m: ~11 mo-1 yr | 10339 |
| | f/m: 1st mate: 9-12 mo | 3571 |
| | f/m: 10-11 mo old | 9916 |
| | f/m: ~1 yr; occasionally yr of birth | 1123 |
| *Sorex cinereus* | f: 4-5 mo | 1474 |
| *Sorex fumeus* | f/m: 1st mate: 8-12 mo | 4277 |
| *Sorex isodon* | m: testes large: 9 mo-1 yr | 10019 |
| *Sorex minutus* | f: 1st mate: 9-12 mo; a few f already 3-5 mo | 3571 |
| | f: 1st birth: 9-10 mo | 9175 |
| | m: ~4 g; min 3.4 g | 4001 |
| | m: > 4 g | 1217 |
| | f/m: not 1st season | 1217 |
| *Sorex mirabilis* | f/m: 11 mo | 8079 |
| *Sorex palustris* | f: 3+ mo | 1474 |
| | f/m: 1st mate: occasionally 4 mo | 2001 |
| *Sorex vagrans* | m: 1st mate: 8.4 g | 5373 |
| | f: 1st mate: 7.7g | 5373 |
| | f: 1st preg: yr of birth | 5373 |
| | f: 1 yr | 11230 |
| *Suncus etruscus* | f/m: prob 1 yr | 3373 |
| *Suncus murinus* | f: large follicles: 3 wk | 5506 |
| | f: 1st cl: 20 g; 50% preg or lactating 32 g | 1331 |
| | f: 1st mate: 30 d | 2737 |
| | f: 1st mate: 5 mo | 7524 |
| | f: 1st mate: before fully adult | 8501 |
| | f: 1st preg: 27 g | 4406 |
| | f: 1st repro: 60 d | 2737 |
| | m: spermatogenesis: 5 wk | 5506 |
| | m: testes anatomy: 44.5 g | 1331 |
| | m: testicular sperm: 27 d | 4476 |
| | m: testicular sperm: 31 d | 2737 |
| | m: 1st sire: 51 d | 2737 |
| | m: 50% fertile 31.5 g, 95 % fertile 46.1 g | 4406 |
| *Suncus varilla* | m: epididymal sperm: >9 mo | 6658 |
| **CYCLE LENGTH** | | |
| *Blarina brevicauda* | estrus: 2-4 d | 8299 |
| *Sorex araneus* | estrus: < 1 d | 2045 |
| **GESTATION** | | |
| *Blarina brevicauda* | 15 d | 7333 |
| | 18-20 d | 8299 |
| | ≥21 d; n1 | 4266 |
| *Crocidura flavescens* | r15-35; n12 | 2955 |
| *Crocidura hirta* | 18 d | 2449, 5759, 10158 |
| | ~18 d | 7167a |
| *Crocidura leucodon* | est 20-35 d | 11391 |
| | 31 d | 3455 |
| | 31-33 d | 3571, 8798 |
| *Crocidura russula* | 28-31 d | 3571 |
| | 28.5 d; mode 29; r24-32 | 4600, 4601, 4604 |
| | 30 d | 11316 |
| *Crocidura suaveolens* | 26.8 d; mode 26-27; r26-29 d; n13 | 11313 |
| | 28 d | 9253 |
| *Cryptotis parva* | 12-16 d | 8798 |
| | r21-23 d; n5 | 2004, 7409, 7410 |
| *Neomys anomalus* | 26 d | 7272 |
| *Neomys fodiens* | 20.5 d; r19-22; n8 lt, 5 f; pairing to birth | 7273, 7276 |
| | 20 d; r20-22 | 11316 |
| | ~24 d; non-lactating f | 8757 |
| | est 24 d | 3630 |

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|---|---|---|
| *Sorex araneus* | 13-19 d | 1210 |
| | 20 d | 2437 |
| | 21 d; n1 | 2156 |
| | 23-28 d | 11316 |
| *Sorex coronatus* | 24.3 d; r24-25; n3 | 3841 |
| *Sorex haydeni* | 19-22 d | 1871a |
| *Sorex fumeus* | ~3 wk | 4277 |
| *Sorex minutus* | ~3 wk | 3571 |
| | ~22 d | 9175 |
| | ~4 wk | 8049 |
| *Sorex vagrans* | 20 d | 5373 |
| *Suncus etruscus* | est 27-28 d; pairing to birth, min ibi | 3373 |
| | 27.5; 4-5 d longer with lactation | 11315 |
| *Suncus murinus* | implant: 7 d | 2737 |
| | implant: 7-8 d | 4480 |
| | placentation: 12 d | 4480 |
| | 28-32 d | 9914 |
| | 29.6 d; se0.1; n53 | 4477 |
| | 30-31 d | 7524 |
| | 30.2 d; mode 30-31; r28.5-31.5; n20 | 2737 |
| | 30.3 d; se0.1; n156 | 4477 |
| **LACTATION** | | |
| *Blarina brevicauda* | leave nest: 12 d | 7333 |
| | solid food: 21 d | 1028 |
| | wean: mode 25 d; r22-30 | 1028 |
| | wean: 22 d | 6882 |
| *Crocidura bicolor* | solid food: min 11 d; ~15 d | 335 |
| | solid food: 19 d | 2589 |
| | nurse: 15 d | 335 |
| *Crocidura hirta* | wean: 15-20 d | 7167a |
| | wean: 18 d | 5759 |
| *Crocidura leucodon* | solid food: 16 d | 11391 |
| | wean: 18-19 d | 11391 |
| | wean: 26 d | 8798 |
| | wean: 16-26 d | 3571 |
| *Crocidura russula* | solid food: 2 wk | 4602, 4603 |
| | wean: 22-24 d | 4602, 4603 |
| | independent: 6 wk | 8049 |
| | wean: ~16 d | 3571 |
| *Crocidura suaveolens* | independent: 6 wk | 8049 |
| *Cryptotis parva* | wean: 18-21 d | 2004, 7409, 7410 |
| *Myosorex varius* | wean: 20-22 d | 673 |
| *Neomys anomalus* | wean: 29-30 d | 7272 |
| *Neomys fodiens* | solid food; 27.8 d; sd1.0; r27-30 | 7273, 7275, 7276, 7276a |
| | nurse: 38 d; r34-41; n7 lt | 7273, 7275, 7276, 7276a |
| | wean: est 21 d | 3630 |
| | wean: 4 wk | 2156 |
| | independent: 6 wk | 8049 |
| | wean: ~37 d | 8757 |
| | wean: 6 wk; r5-6 wk | 3571 |
| *Notiosorex crawfordi* | independent: 40 d | 4901 |
| *Sorex araneus* | den emergence: 17 d | 2437 |
| | den emergence: 18-19 d | 12110 |
| | den emergence: 3 wk | 3571 |
| | feeding by themselves: 22 d | 2437 |
| | wean: 21 d | 11314 |
| | wean: 4 wk | 2156 |
| *Sorex cinereus* | wean: 20-27 d | 3423 |
| *Sorex minutus* | wean: 22 d | 9175 |
| | wean: 23 d | 2156 |
| *Sorex monticolus* | wean: <3 wk | 1871a |
| *Sorex vagrans* | wean: ~16-25 d | 5373 |
| *Suncus etruscus* | wean: 19-21 d | 3373 |
| | wean: 20 d | 11315 |

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| *Suncus murinus* | solid food: 20 d | 2736 |
| | wean: 2+ wk | 7524 |
| | wean: ~20 d | 2736 |
| | wean: 20 d | 9914 |
| **INTERLITTER INTERVAL** | | |
| *Crocidura hirta* | 59 d; n1 | 7167a |
| *Crocidura russula* | 28.5 d; mode 29; r24-32 | 4600, 4601, 4602, 4604 |
| *Neomys fodiens* | 20 d-4 mo | 11316 |
| *Sorex araneus* | 28-43 d | 11316 |
| | 43-44 d | 9712 |
| *Sorex nanus* | 1-2 mo? | 4894 |
| *Suncus murinus* | 56.2 d; r30-101; n12; yg survive | 2737 |
| **SEASONALITY** | | |
| *Anourosorex squamipes* | testes: large June-Aug; small Nov-Dec; Taiwan | 148 |
| *Blarina brevicauda* | sperm: none Nov-early Jan; e US | 8299 |
| | testes: large end Apr-early Sept; QUE, Canada | 11956 |
| | mate: est by 10 Feb; PA | 1831 |
| | mate: est Mar-Aug; se MAN, Canada | 1426 |
| | preg f: Apr-May; n4; NE | 5405b$_1$ |
| | preg f: May; n1; MN | 495 |
| | birth: by end Feb; PA | 1831 |
| | birth: Mar-May, Aug-Sept; US | 1474 |
| | birth: Apr-Sept; e US | 8299 |
| | birth: Apr-Sept; PA | 9164 |
| | birth: peak May-June; NY | 2261 |
| | birth: spring, late summer; NY | 4266 |
| | lact f: May, July-Aug; NE | 5405b$_1$ |
| | lact f: Oct; FL | 7498a |
| | repro: peak Mar-July, 2nd peak Sept; IL | 3743 |
| | repro: Apr, May (and Nov?); NY | 7235 |
| *Blarina carolinensis* | preg f: Mar-Apr, July; n3; NC | 1290 |
| *Crocidura* spp | preg f: yr rd except May, July, Dec; Zambia | 9893 |
| *Crocidura allex* | preg f: Sept; n1; Aberdare Mt, Kenya | 4924 |
| *Crocidura bicolor* | preg f: Nov-Dec; s Africa | 7167a |
| | birth: Oct; few data; n Zimbabwe | 335 |
| | lact f: Apr; n1; n Zimbabwe | 331 |
| *Crocidura bottegi* | preg f: Mar; n1; Nigeria | 4354a |
| *Crocidura crossei* | preg f: Apr-May, Oct-Nov; Nigeria | 4354a |
| | preg f: May, July-Aug, Oct; Nigeria | 5190a |
| *Crocidura cyanea* | birth: est warm, wet summer; S Africa | 10158 |
| | yg found: Nov-Jan; s Africa | 7167a |
| *Crocidura flavescens* | preg f: Jan; n1; Kenya, Uganda | 4924 |
| | preg f: Jan; n1; Namibia | 9945 |
| | preg f: Jan; n2; captive | 6828a |
| | preg f: Oct-Nov; Nigeria | 5190a |
| | preg f: yr rd; Uganda, Rwanda, Burundi | 2575 |
| | young found: most mo; e Africa | 5759 |
| *Crocidura foxi* | preg f: July; n1; Nigeria | 4354a |
| *Crocidura fuliginosa* | preg/lact f: Mar-Apr, Sept-Nov; n4; Malaysia | 7161 |
| | juveniles present: Aug; Malaysia | 7161 |
| *Crocidura gracilipes* | sexually active f: Mar, June, Nov; Uganda, Rwanda, Burundi | 2575 |
| | lact f: Aug; n1; e Africa | 5759 |
| *Crocidura grandiceps?* | preg f: Mar, May-June, Sept; Nigeria | 5190a |
| *Crocidura hirta* | preg f: Nov; n1; Namibia | 9945 |
| | preg f: Nov-May; Transvaal, S Africa | 8965 |
| | preg f: Dec-Feb; n5; s Africa | 7167a |
| | birth: Jan-Apr; n4; captive | 7167a |
| | birth: Sept-May; S Africa | 10158 |
| | lact f: Mar-May; s Tanzania | 187 |
| | lact f: Apr; n1; s Africa | 7167a |
| *Crocidura horsfieldi* | mate: est July-Aug; Sri Lanka | 8501 |
| | preg f: Apr; n3; Nepal | 7382 |
| *Crocidura jacksoni* | preg f: Nov, June; n2; Kenya | 4924 |

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|---|---|---|
| *Crocidura leucodon* | birth: spring-fall; Germany | 3571 |
| | repro: not seasonal; Germany | 8798 |
| *Crocidura luna* | preg f: Mar-Apr; Zambia | 7167a |
| | preg f: Apr; n1; s Africa | 7167a |
| | lact f: Apr; n1; s Africa | 7167a |
| *Crocidura mariquensis* | mate: end dry season-end rains; Zambia | 9893 |
| | preg f: Aug-Apr; Botswana | 10154 |
| | preg f: Aug-Dec, Feb-Apr; S Africa | 10158 |
| | preg/lact f: Jan-Feb, Oct; Transvaal, S Africa | 8965 |
| *Crocidura russula* | mate: Feb-Sept, but low Aug; Spain | 6571 |
| | mate: end Feb-Oct; Channel I, UK | 955 |
| | mate: yr rd; Germany | 3571 |
| | mate: yr rd, peak Mar-May; captive | 4600 |
| | preg f: Feb-Sept; Spain | 6574 |
| | preg f: Mar-Sept; Italy | 5485 |
| | preg f: peak Apr, low Dec; lab | 4601 |
| | birth: yr rd; lab | 11316 |
| *Crocidura suaveolens* | birth: Apr-Sept; lab | 11313 |
| | birth: almost yr rd; e Europe, n Asia | 8049 |
| *Crocidura turba* | preg f: July; n1; ne Zaire | 2372 |
| *Crocidura zarudnyi* | preg f: May; n1; Iran | 2404a |
| *Cryptotis magna* | preg f: May, Oct; Oaxaca, Mexico | 9177 |
| *Cryptotis mexicana* | preg f: Dec; n1; Veracruz, Mexico | 4228 |
| | lact f: Oct; Veracruz, Mexico | 4228 |
| *Cryptotis parva* | mate: Mar-Nov; north; yr rd; south; US | 4279 |
| | preg f: Feb, May, Sept; n3; NC | 1290 |
| | preg f: June, Aug, Oct; n3; FL | 6303 |
| | preg f: Oct; n1; KS | 1812a |
| | preg/lact f: Apr-Oct; FL | 6303 |
| *Diplomesodon pulchellum* | birth: Apr-Aug; USSR | 4645 |
| *Megasorex gigas* | lact f: June; n1; Guerrero, Mexico | 368 |
| *Myosorex cafer* | preg f: Oct-Nov; n3; Zimbabwe | 10158 |
| *Myosorex norae* | preg f: Oct; n1; Kenya | 5759 |
| | preg f: Oct; n1; Kenya | 4924 |
| *Myosorex polulus* | preg f: Oct; n2; Kenya | 4924 |
| *Myosorex varius* | testes: small May-June; S Africa | 9329 |
| | mate: Sept-Mar; S Africa | 9329 |
| | lact f: Sept-Oct, Feb, Apr; n9; Transvaal, S Africa | 8965 |
| *Neomys anomalus* | mate: Mar-Apr?; Germany | 3571 |
| | birth: June-Aug; n4; captive | 7272 |
| *Neomys fodiens* | testes: large Apr-May; lab | 8757 |
| | mate: Apr-May; Baltic area, Europe | 8049 |
| | mate: Apr-Sept; central Europe | 3571 |
| | mate: mid Apr-Sept, peak May-June; lab | 8757 |
| | birth: May-June; Europe, w USSR | 4645 |
| | birth: May-Oct; central Europe | 3571 |
| | repro: low Aug; lab | 11316 |
| | repro active f: end Mar-Oct; Germany | 3630 |
| | repro: Mar-?; Germany | 10347 |
| | sexually active: Apr-Sept; Poland | 1123 |
| *Notiosorex crawfordi* | preg f: Apr; n1; CA | 369 |
| | preg f: Nov; n2; OK | 529 |
| | birth: summer; AZ, CA | 4901 |
| *Scutisorex somereni* | sexually active f: Jan, Mar-Apr, June-Aug, Dec; central Africa | 2575 |
| | lact f: May; n3; Uganda, Zaire | 5759 |
| *Sorex alpinus* | birth: May, Aug; Germany | 3571 |
| *Sorex araneus* | sperm: mid Mar-early Nov; Wales, England, UK | 1211 |
| | spermatogenesis: end Feb-early Mar; s Britain | 2045 |
| | testes: large Feb-July; UK | 2156 |
| | testes: large end Feb-Apr; n Finland | 4577 |
| | testes: large Apr-Sept; England, UK | 1838 |
| | uteri: large Apr-July; England, UK | 1838 |
| | uteri: large May-July; UK | 2156 |
| | estrus: mid-end Apr, mid-end June; Germany | 9159 |

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|---|---|---|
| | mate: Mar-?; England, UK | 1427 |
| | mate: Mar-Apr; Europe, USSR | 4645 |
| | mate: est Apr/May-?; Finland | 10017 |
| | preg f: Apr-Aug; n Finland | 4577 |
| | preg f: Apr-early Sept; e Germany | 10339 |
| | preg f: Apr-Sept; Poland | 2436 |
| | preg f: peak May; few data; Britain | 8372 |
| | preg f: May-June; Germany | 10344 |
| | preg f: May-Oct; Great Britain | 1210 |
| | preg f: May-Oct, none Nov; Poland | 10677 |
| | preg f: May-Oct; Poland | 2436 |
| | preg f: May-Oct; Wales, Scotland, UK | 1210 |
| | birth: (Apr) May-Oct; Germany | 3571 |
| | birth: end Apr-Aug; England, UK | 9916 |
| | birth: end May-mid Sept; n9; lab | 11314 |
| | birth: May-Oct; Wales, Scotland, UK | 1210 |
| | birth: mid May-Oct; Poland | 10677 |
| | birth: spring-summer; s Britain | 2045 |
| | sexually active: Apr-Oct; Poland | 1123 |
| *Sorex arcticus* | active testes: Feb-June; WI | 1912 |
| | mate: est Mar-Aug; se MAN, Canada | 1426 |
| | preg f: May, July; n2; NS, Canada | 10144 |
| | preg f: Apr, June; n2; MN | 495 |
| | preg f: Sept; n1; none end July-early Sept; AK | 716 |
| | preg/lact f: Apr-Sept; MN | 506 |
| | birth: early-mid May; BC, Canada | 2088 |
| *Sorex cinereus* | testes: large spring; few data; s QUE, Canada | 11956 |
| | estrus: spring; few data; s QUE, Canada | 11956 |
| | mate: est May-Aug; se MAN, Canada | 1426 |
| | preg f: Jan, Apr-May, Sept; N America | 1474 |
| | preg f: Apr, June; n2; MN | 495 |
| | preg f: mid June-mid July; n12; SD, WY | 10954 |
| | preg f: July; n1; se MT | 6166 |
| | preg f: July-Aug; n2; YUK, Canada | 12049 |
| | preg f: none mid July-early Sept; n39; AK | 716 |
| | lact f: July; n1; ND | 3703 |
| | lact f: July; n1; NE | 5405b, |
| *Sorex coronatus* | preg f: Feb-Sept; n Iberian Peninsula | 6573 |
| | preg f: May-July; ne Iberian Peninsula | 6572 |
| *Sorex cylindricauda* | preg f: Sept; n1; Nepal | 7382 |
| *Sorex dispar* | testes: large Apr-Aug; VT | 5799 |
| | preg f: May; n2; PA | 9111a |
| | preg f: Aug; n1; NY | 10683a |
| | preg/lact f: May-Aug; VT | 5799 |
| | birth: May; ne US | 1474 |
| *Sorex fontinalis* | f fertile: end Feb-?; PA | 9164 |
| | preg f: end Mar-Sept; PA | 9164 |
| *Sorex fumeus* | testes: large Apr-Aug; NY | 4277 |
| | mate: mid Apr-Aug; QUE, Canada | 11956 |
| | preg f: Mar-Apr; TN, NC | 8155 |
| | preg f: May; n1; NS, Canada | 10144 |
| | birth: Apr-Oct; e US, Canada | 1474 |
| *Sorex gaspensis* | testes: large: end Apr-Aug; NB, Canada | 5799 |
| *Sorex hoyi* | preg f: July; n1; IA | 6554 |
| | preg f: July; n1; IA | 9700 |
| | lact f: July-Aug; n2; WI | 6554 |
| | lact f: Aug; n1; CO | 6554 |
| | lact f: Aug; n1; AK | 6554 |
| *Sorex isodon* | preg f: mid May-early Oct; e Finland | 10019 |
| *Sorex longirostris* | preg f: Apr-Sept, peak Apr; se US | 3512 |
| | birth: Apr; e US | 1474 |
| *Sorex merriami* | preg f: Apr, May, July; WA | 5352 |
| | large mammary glands: Mar, Oct; WA | 5352 |

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|---|---|---|
| *Sorex minutus* | spermatogenesis: Mar-Aug; Wales | 1217 |
| | mate: Apr-Aug; mid Europe | 3571 |
| | mate: Apr-Oct; Poland | 1123 |
| | mate: May?; USSR | 8049 |
| | preg f: mid Apr-early Oct, peak June; Wales | 1217 |
| | preg f: end Apr-early May, early July; Germany | 9159 |
| | preg f: May; n1; Nepal | 7382 |
| | preg f: May-June; few data; Britain | 8372 |
| | preg, lact f: Apr-Oct; Poland | 2436 |
| | birth: ?-Aug; Pakistan | 9175 |
| | birth: May-Sept; mid Europe | 3571 |
| | birth: June; USSR | 8049 |
| | birth: June-Oct; Ireland | 4001 |
| *Sorex monticolus* | preg f: June, Aug; n2; WY | 1871a |
| | birth: July; few data; nw N Am | 1474 |
| *Sorex nanus* | birth: July-Sept; WY | 1871a |
| | birth: end July-early Sept; alpine US | 4894 |
| *Sorex ornatus* | repro: end Feb-Oct; CA | 8156 |
| *Sorex pacificus* | preg f: Mar, May-Sept; CA | 1614 |
| | preg f: mid Apr-Aug, Nov; OR | 1614 |
| *Sorex palustris* | sperm present; Jan-July; MT | 2001 |
| | preg f: Mar; WY | 1871a |
| | birth: end Feb-June; n US | 1474 |
| *Sorex raddei* | mate: Mar; Turkey | 10274 |
| | birth: mid-end May; Turkey | 10274 |
| *Sorex trowbridgii* | mate: Feb-early June; CA | 5244 |
| | mate: Feb-Oct; OR | 3623 |
| | birth: Mar-May, occasionally July; w US | 1474 |
| | birth: end Apr/early May-?; BC, Canada | 2088 |
| *Sorex vagrans* | preg f: Apr-early Aug; MT | 1910 |
| | preg f: mid June-mid Aug; few data; CO | 11230 |
| | preg f: Sept, mid Feb-early June; CA | 5373 |
| | birth: Feb-May; Oct-Nov; nw US | 1474 |
| | birth: peak end Feb-early June, sm peak fall; CA | 5373 |
| | birth: Mar-July; WY | 1871a |
| | birth: Mar-summer; s BC, Canada | 2088 |
| | birth: spring; w US | 5146 |
| *Soriculus caudatus* | preg f: May, Aug; n8; Nepal | 7382 |
| *Soriculus fumidus* | testes: large Mar-Apr; Taiwan | 148 |
| *Soriculus nigrescens* | preg f: May, July-Aug, Oct; n9; Nepal | 7382 |
| *Suncus etruscus* | preg/lact f: May-Sept; France | 3373, 3374 |
| | birth: yr rd, peak Aug-Oct; Pakistan | 9175 |
| | repro: Apr-May; lab | 11315 |
| *Suncus lixus* | preg f: Jan; n1; Zambia | 328a |
| *Suncus murinus* | epididymal sperm: yr-rd; w Rajasthan, India | 8903 |
| | sperm: peak Jan-June; Malaysia | 4406 |
| | sperm: yr rd; Pakistan | 494 |
| | mate: yr rd, peak June-Sept; Japan | 7524 |
| | preg f: peak Jan-Mar, Apr-June; Burma | 1331 |
| | preg f: Feb-Aug; Pakistan | 6728 |
| | preg f: Mar-Sept, peak June-July; w Rajasthan, India | 8903 |
| | preg f: Apr, June-July, Sept; n7; Nepal | 7382 |
| | preg f: June, July, Dec; n3; Sri Lanka | 8501 |
| | preg f: yr rd, peak Mar-Apr, July-Aug; Zhejiang, China | 11999a |
| | preg f: yr rd, peak Apr-June, Oct-Dec; Guam | 572 |
| | preg f: yr rd, peak 42% Oct-Dec; Malaysia | 4406 |
| | preg f: yr rd; w Bengal, India | 6589 |
| | birth: spring & end of monsoon (end Sept); Pakistan | 9175 |
| | young found: yr rd; Guam | 8410a |
| *Suncus varilla* | testes: peak Aug-Feb; S Africa | 6658 |
| | mate: Aug-Mar; S Africa | 6658 |
| | preg f: peak Aug-Oct; S Africa | 6658 |

## Order Insectivora, Family Talpidae

Moles (*Condylura* 35-85 g; *Desmana* 100-220 g; *Galemys* 50-80 g; *Mogera* 87-220 mm; *Neurotrichus* 8-11 g; *Parascalops* 40-85 g; *Parascaptor* 110-115 mm; *Scalopus* 40-140 g; *Scapanulus* 35-38 mm; *Scapanus* 50-170 g; *Scaptochirus* ~140 mm; *Scaptonyx* 65-90 mm; *Talpa* 60-150 g; *Uropsilus* 12-20 g; *Urotrichus* 15 g) are subterranean or semiaquatic insectivores from Eurasia and North America. Males may be larger than females (*Talpa* 3842, 7186, 10341; *Parascalops* 2844).

Copulation may trigger ovulation (*Talpa* 2045, 2423; *Scalopus* 2005). A copulatory plug is present in *Condylura* and *Parascalops* (2847, 2848). The villous, labyrinthine, diffuse, discoid placenta invades the uterine wall (231, 7526, 7563, 7564, 7567, 8736, 10310), has an antimesometrial, superficial implantation into the bipartite or bicornuate uterus (2662, 7526, 7563, 8460, 8736, 8789, 10341), and is absorbed after birth (2423, 6955; cf 8462). Delayed implantation may occur (*Talpa* 486) as well as a postpartum estrus (*Galemys* 8209).

Neonates are blind and naked (2231, 2844, 4268, 4455, 5209, 6088, 11897). Their eyes open at 20-22 d (*Talpa* 2045, 3571, 3842, 8049). Females have 6-8 mammae (63, 1474, 2662, 7621, 8049), several accessory reproductive glands, and a long, penis-like clitoris (3833, 7186, 8049, 8459, 8460, 10046, 11897). A vaginal opening exists only during the breeding season (*Talpa* 63, 6955, 11897; *Scalopus* 2005; *Parascalops* 2844).

Males have abdominal testes (2045, 2846, 2848, 3833). *Neurotrichus* males have ampullary glands (2849), while *Condylura*, *Neurotrichus*, and *Parascalops* males have inguinal and perineal glands (2846, 2848, 2849). The penis is usually decorated (2848, 2849, 8789), although not in *Parascalops*. Both *Parascalops* and *Talpa* have bacula (2846, 3833). The male reproductive system can reach 14% of body mass in *Condylura* (2848, 4268).

Details of anatomy (211, 2005, 2423, 2775, 4105, 4530, 6334, 8678, 8789, 10566, 10567), intersex (1360), and endocrinology (3831, 8049, 8832, 9007, 10626, 10668) are available.

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|---|---|---|
| **NEONATAL MASS** | | |
| *Neurotrichus gibbsii* | 0.67 g; n1 | 2231 |
| *Scalopus aquaticus* | 5.355 g; r5.35-5.36; n2 | 4455 |
| *Scapanus townsendii* | est ~5 g | 6088 |
| *Talpa europaea* | 3 g | 8274 |
| | 3.5 g | 63 |
| **NEONATAL SIZE** | | |
| *Galemys pyrenaicus* | 30-32 mm; large emb | 8209 |
| *Neurotrichus gibbsii* | HB: 21 mm; n1 | 2231 |
| *Scapanus townsendii* | 50 mm | 6088 |
| *Talpa europaea* | 30 mm | 8274 |
| | 33-34 mm | 3834 |
| | 35 mm | 7186 |
| | 35-40 mm | 3842 |
| | 44 mm | 63 |
| **WEANING MASS** | | |
| *Scapanus townsendii* | 60-80 g; 30 d | 6088 |
| *Talpa europaea* | 44.2 g; r33-67; n5, 33 d | 3837 |
| | ~60 g; 3 wk | 2045 |
| **WEANING SIZE** | | |
| *Scapanus townsendii* | ≥115 mm | 6088 |
| **LITTER SIZE** | | |
| *Condylura cristata* | 2 yg; n1 nest | 11667 |
| | 5 emb; n1 f | 11956 |
| | 5 ps; n1 f | 10839 |
| | 5 yg; r3-7; n9 lt | 2851 |
| | 5.4 ps; n9 f | 2851 |
| | 5.44 emb; r2-7; n25 f | 2851 |
| | 5.6 yg; r5-7; n5 nests | 2322 |
| | 6.3 emb; r6-7; n7 f | 4268 |

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|---|---|---|
| *Desmana moschata* | 3-5 | 4645 |
| | 3-5 | 8049 |
| *Galemys pyrenaicus* | 3.6 emb; r1-5; n53 f | 8209 |
| | mode 4 yg; r1-5 | 8461 |
| *Neurotrichus gibbsii* | 2.8 emb, yg; r1-4; n5 lt | 2231 |
| | 4 emb; n1 f | 8825a |
| *Parascalops breweri* | 4 | 2844 |
| | 4-5 emb; max 8 | 4247 |
| *Scalopus aquaticus* | 1 emb; n1 f | 359a |
| | 2-4 emb; n5 f | 1290 |
| | 2 emb; n1 | 8026 |
| | 3 emb; n1 f | 5405b$_1$ |
| | 3 emb; n1 f | 5803 |
| | 3 emb; n1 f | 12006a |
| | 3.9; mode 4 | 8736 |
| | 3.91 emb; mode 4; r3-5; n33 f | 2005 |
| | mode 4; r2-5 | 5209 |
| | 4-5 emb | 359a |
| | 4 ps; n1 f | 9164 |
| | 4.33 cl; r3-7; n15 f | 2005 |
| *Scapanus latimanus* | 2-5 | 1474 |
| *Scapanus orarius* | 2-4 emb | 3816 |
| | 2-5 | 2088 |
| *Scapanus townsendii* | 2.5 emb; mode 3; r1-3; n18 f | 6088 |
| | 2.87 yg; mode 3; r1-4; n94 nests | 6088 |
| | 3.5 emb, yg; mode 3; r2-6; n8 f | 7490 |
| *Talpa altaica* | est 4.6; r3-6 | 5463 |
| *Talpa europaea* | 2-6 | 3834 |
| | 2-6 emb | 3833 |
| | 2.99 emb text, 3.0 emb table; r2-4; n10 f | 11255 |
| | 3-4 | 6955 |
| | 3-4 | 2423 |
| | mode 3-7; r3-9 | 3571 |
| | 3 emb; 1 f | 1590 |
| | 3.12 emb; r2-4; n17 f | 6573a |
| | 3.5; se0.17; mode 4; n48 lt | 3842 |
| | 3.63; r1-7; n60 lt | 63 |
| | 3.65 emb; r1-6; n26 f | 10344 |
| | 3.80; se0.11; mode 4; n45 lt | 6216 |
| | 3.82 emb; ±0.13; r2-6 | 3836 |
| | 4 emb; n1 f | 7902 |
| | 4.54 emb; r3-7; n46 f | 10341 |
| | 4.8; mode 4 90%; r2-7; n68 lt | 545 |
| | 5.08; se0.39; mode 5; n13 lt | 635 |
| | 5.73; se0.06; mode 6; n11 lt | 7357 |
| *Urotrichus talpoides* | 3-4 | 11051 |
| | 3.5 emb; mode 4; r2-5; n19 f | 12023 |
| | 3.6; n8 lt | 5168 |
| | 3.75 emb; se0.44 | 5137 |
| | 3.9 emb; mode 3; r3-5; n7 f | 11050 |
| **SEXUAL MATURITY** | | |
| *Condylura cristata* | f/m: ~10 mo | 1474 |
| *Parascalops breweri* | f/m: 10 mo | 2844 |
| *Scalopus aquaticus* | f/m: 1st mate: 10 mo | 1871a |
| | f/m: 1st mate: 1 yr | 1474 |
| | f/m: not yr of birth | 2005 |
| *Scapanus orarius* | f/m: 9-10 mo | 4456 |

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|---|---|---|
| *Talpa europaea* | f: 1st mate: 1 yr | 10341 |
| | f: 1st conceive: 8 mo | 3834 |
| | f: 1st birth: spring of yr after birth | 2045 |
| | f: 1st birth: 12 mo | 3571 |
| | f: 1 yr | 6573a |
| | m: 1st mate: 1 yr | 7186 |
| | m: 1 yr | 3842 |
| | m: 1 yr | 6573a |
| *Urotrichus talpoides* | f/m: est 1 yr | 5137 |
| **CYCLE LENGTH** | | |
| *Talpa europaea* | estrus: est 20-30 h | 2045 |
| | <1 mo | 6955 |
| **GESTATION** | | |
| *Desmana moschata* | 45-50 d | 4645 |
| *Galemys pyrenaicus* | ~30 d | 8209 |
| *Neurotrichus gibbsii* | >15 d; capture to birth | 2231 |
| *Parascalops breweri* | est 4-6 wk | 2844 |
| *Scalopus aquaticus* | est ~4 wk | 2005 |
| | 40-45 d | 1871a |
| | ~6 wk | 359a |
| | ~6 wk | 1474 |
| | ~45 d | 5210 |
| *Talpa altaica* | 9-10 mo | 1122 |
| *Talpa europaea* | ~3 wk; field data | 10341 |
| | 30-40 d | 4645 |
| | ~4 wk | 6955 |
| | est 4+ wk | 63 |
| | 1 mo | 2423 |
| | 30 d | 3833, 3834 |
| | 40 ? d; r3-4 ? wk | 3571 |
| **LACTATION** | | |
| *Condylura cristata* | independent: 3 wk | 1474 |
| *Desmana moschata* | independent: 40 d | 4645 |
| *Parascalops breweri* | leave nest: ~1 mo | 1474 |
| *Scalopus aquaticus* | independent: ~1 mo | 1474 |
| | independent: 5-6 wk | 1871a |
| *Scapanus townsendii* | leave nest: 30 d | 6088 |
| *Talpa europaea* | wean: 1 mo | 8274 |
| | wean: 4-5 wk | 2423 |
| | wean: 33 d | 3837 |
| | independent: 4-6 wk | 3571 |
| | independent: 5-6 wk | 4645 |
| | independent: 5-6 wk | 8049 |
| | wean: 3 wk | 6955 |
| *Urotrichus talpoides* | independent: 4 wk | 5168 |
| **LITTERS/YEAR** | | |
| *Condylura cristata* | 1 | 4268 |
| | min 2 | 7235 |
| *Desmana moschata* | est 2 | 8049 |
| *Neurotrichus gibbsii* | >1 | 1474 |
| *Parascalops breweri* | 1 | 2844 |
| | 1; possibly 2 | 1474 |
| *Scalopus aquaticus* | 1 | 2005 |
| | est 1 | 5209 |
| *Scapanus latimanus* | 1 | 1474 |
| *Scapanus orarius* | 1 | 1474 |
| *Scapanus townsendii* | 1 | 1474, 2088, 9575 |
| *Talpa europaea* | 1 | 6955 |
| | 1 | 63 |
| | 1 | 10341 |
| | 1 | 2423 |
| | 1+; mode 1; 3% 2; r1-2 | 3842 |
| | mode 1; 3% 2; r1-2 | 7186 |
| *Urotrichus talpoides* | 1-2 | 5137 |

**SEASONALITY**

| | | |
|---|---|---|
| *Condylura cristata* | mate: end Mar-Apr; NY | 2847, 2848 |
| | birth: end Mar-early Aug; NY | 2851 |
| | birth: Apr-June; NC, ne US, e Canada | 1474 |
| | birth: Apr-June, peak early May; NY | 4268 |
| | preg f: end Mar-early June; NY | 4268 |
| *Desmana moschata* | birth: yr rd, peak spring; USSR | 4645 |
| *Galemys pyrenaicus* | m fertile: Nov-May; France | 8461 |
| | preg f: Feb-June; France | 8459, 8461, 8462 |
| *Neurotrichus gibbsii* | m fertile: Feb; n1; WA | 2231 |
| | preg f: Feb; n1; WA | 2231 |
| | preg f: Apr; n1; BC, Canada | 8825a |
| | birth: Mar; BC, Canada | 2088 |
| | lact f: Sept; n2; WA | 2231 |
| | lact f: Sept; BC, Canada | 2088 |
| *Parascalops breweri* | mate: end Mar-early Apr; NH | 2844 |
| | birth: early May; ne US, ne Canada | 1474 |
| *Scalopus aquaticus* | mate: Mar-Apr; s WI | 8736 |
| | mate: end Mar-early Apr; WI | 2005 |
| | mate: Feb-early Mar; MO | 2005 |
| | preg f: Feb; n1; TX | 12006a |
| | preg f: Mar-May; n3; NC | 1290 |
| | preg f: May; n1; NE | 5405b$_1$ |
| | birth: Mar; south; May; north; central, e US | 1474 |
| | birth: Mar-Apr, n US; earlier s US | 5209 |
| *Scapanus latimanus* | birth: Mar-Apr; US | 1474 |
| *Scapanus orarius* | mate: Jan-early Mar; BC, Canada | 3816 |
| | birth: Mar-Apr; BC, Canada | 3816 |
| *Scapanus townsendii* | m fertile: Feb; US | 1474 |
| | mate: Feb; BC, Canada | 2088 |
| | birth: Mar-Apr; US | 1474 |
| | birth: mid-end Mar; WA | 9575 |
| *Talpa altaica* | implant: spring; Siberia | 1122 |
| | birth: spring; Siberia | 1122 |
| *Talpa europaea* | spermatogenesis: Jan-Mar; Austria | 10668 |
| | spermatogenesis: Jan-mid Apr; Europe | 2075 |
| | spermatogenesis: peak Feb-Mar, low May; Britain | 2045 |
| | spermatogenesis: Feb-May; England, UK | 6543 |
| | spermatogenesis: peak Mar, none July-Jan; England, UK | 10567 |
| | spermatogenesis: Dec; location not given | 9007 |
| | epididymal sperm: Mar-Apr, none May; Austria | 10668 |
| | testes: large Jan-Feb, small May-June; France | 3832 |
| | testes: enlarged Feb; Germany | 10341 |
| | testes: large Feb-Mar, small rest of yr; England, UK | 3842 |
| | testes: large Feb-Apr; Britain, Netherlands | 7186 |
| | testes: large Feb-Apr; England, UK | 63 |
| | testes: large Feb-Apr; England, UK | 6543 |
| | testes: large end Feb-early Apr, small Aug; Austria | 10668 |
| | testes: large Mar, small Nov; Lancashire, England, UK | 8832 |
| | testes: large early Apr; central, s Ural, USSR | 8274 |
| | testes/acc gland: large mid Mar-end Apr; England, UK | 2423 |
| | vaginal clefts open: mid Apr; central, s Ural, USSR | 8274 |
| | ovarian medullary cords: max Mar-Apr; France | 3832 |
| | ovary: quiescent June-Feb; France | 3831 |
| | mate: mid Feb-mid Mar; France | 3831, 3833, 3834 |
| | preg f: Jan-Apr; Spain | 6573a |
| | preg f: Feb-June; England, UK | 3836 |
| | preg f: mid Feb-?; Spain | 11255 |
| | preg f: Mar-June; Germany | 10341, 10344 |
| | preg f: Aug; n1; odd, usually none after July; Germany | 7902 |
| | preg/lact: Feb-June; France | 3831 |
| | birth: end Mar-Apr; Austria | 10668 |
| | birth: mid-end Apr; Britain, Netherlands | 7186 |
| | birth: mid Apr-May; central European Russia, w Siberia | 8049 |

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| | birth: mid Apr-mid May; sw European USSR | 4645 |
| | birth: mid Apr-June, peak May; Ukraine w of Dnieper | 7357 |
| | birth: mid Apr-June; England, UK | 63 |
| | birth: end Apr-early May; England, UK | 6955 |
| | birth: May-June; Britain | 2045 |
| | birth: end May-early June, 25% preg summer; central, s Ural, USSR | 8274 |
| | lact f: Feb-June; Spain | 6573a |
| *Talpa micrura* | lact f: Mar; n1; Malaysia, Cameron Highlands | 7161 |
| *Urotrichus talpoides* | active cauda epididymis: Feb-May; Chiba, Japan | 5168 |
| | testes: large Jan-May; Fukucka, Japan | 12023 |
| | testes: large Mar, small Oct; Japan | 11051 |
| | mate: Feb-May, small peak Aug-Sept; Niigata, Honshu, Japan | 5137 |
| | preg f: mid Jan-?; Fukucka, Japan | 12023 |
| | preg f: Mar-Apr; Chiba, Japan | 5168 |
| | preg f: Mar-June; Japan | 11051 |
| | lact f: mid Feb-?; Fukucka, Japan | 12023 |
| | lact f: end Apr-mid June; Japan | 11051 |